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Pollack et al.

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(54) **DEVICES AND METHODS FOR TARGETED DELIVERY OF A SUBSTANCE**

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

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(65) **Prior Publication Data**

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Related U.S. Application Data

(60) Provisional application No. 62/796,338, filed on Jan. 24, 2019, provisional application No. 62/700,768, filed on Jul. 19, 2018.

(51) **Int. Cl.**

A61B 17/00 (2006.01)

A61B 18/18 (2006.01)

(Continued)

(52) **U.S. Cl.**

CPC **A61B 17/00491** (2013.01); **A61B 18/18** (2013.01); **A61F 9/00727** (2013.01);

(Continued)

(58) **Field of Classification Search**

CPC **A61F 9/0008**; **A61F 9/00727**; **A61B 2017/00495**; **A61B 17/00491**;

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Primary Examiner — Elizabeth Houston

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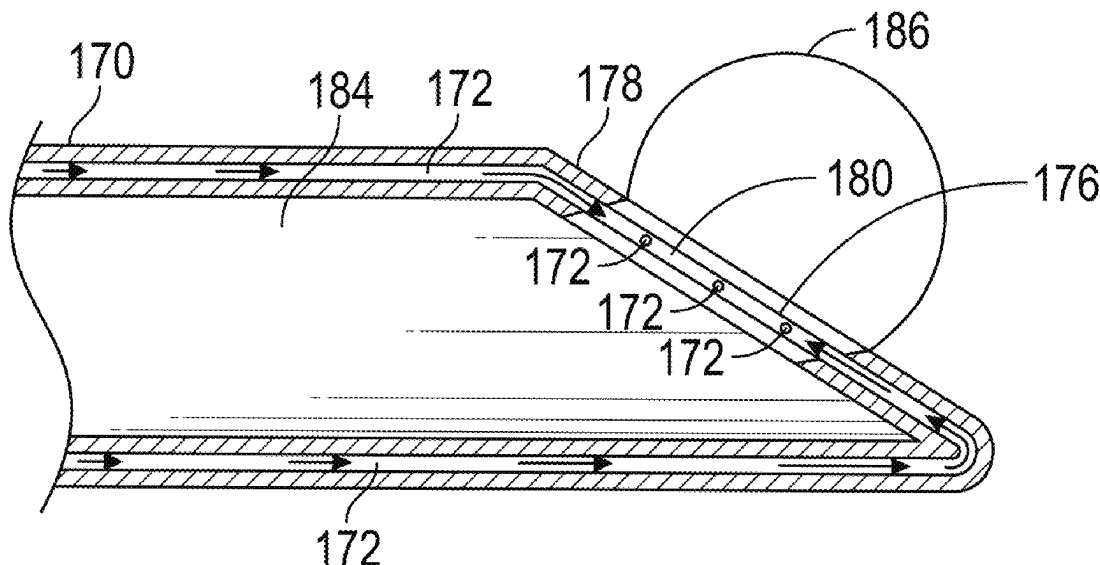
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(57)

ABSTRACT

A device for applying a bubble of a substance to a tissue surface, the device comprising a cannula, a distal tip at the distal end portion of the cannula, the distal tip having a bubble support surface and an exit port extending through the bubble support surface, an expansion fluid passageway extending through the cannula to the exit port, a source of an expansion fluid and an actuator therefor. In some arrangements, the distal tip can be configured to support a layer of the substance thereon over the distal port and the device can be configured such that the advancement of the expansion fluid from the fluid source through the exit port causes at least one bubble of the substance to form on the distal tip, wherein the at least a portion of the bubble can be transferred to the tissue surface to treat a defect on the tissue surface.

42 Claims, 17 Drawing Sheets



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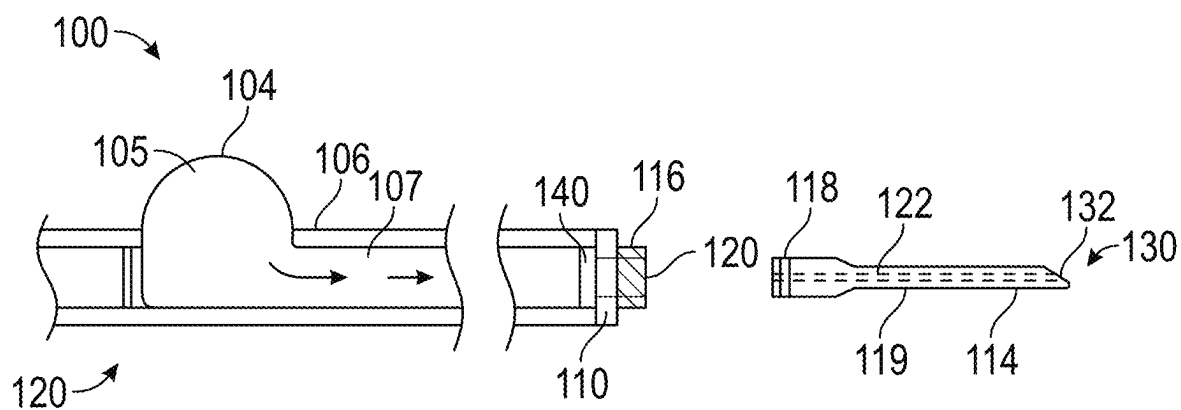


FIG. 1

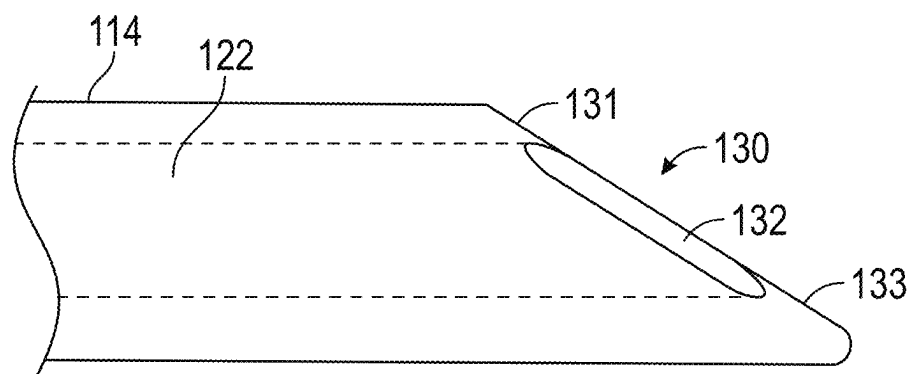


FIG. 2

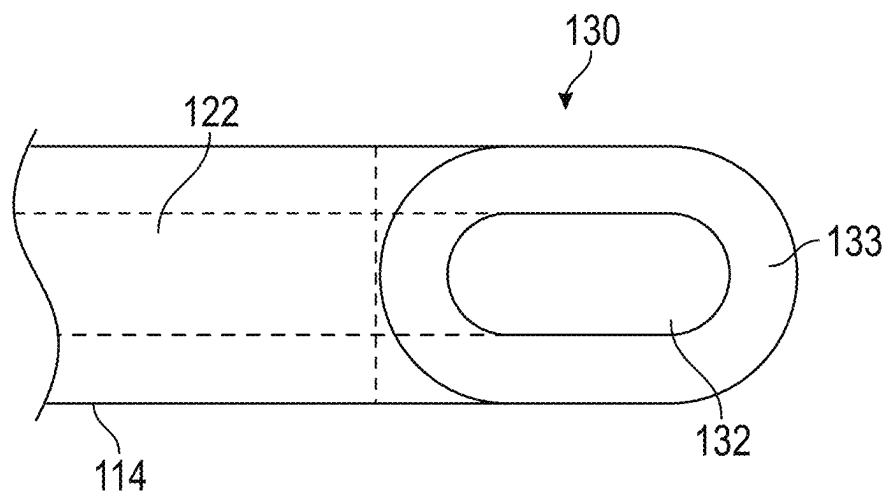


FIG. 3

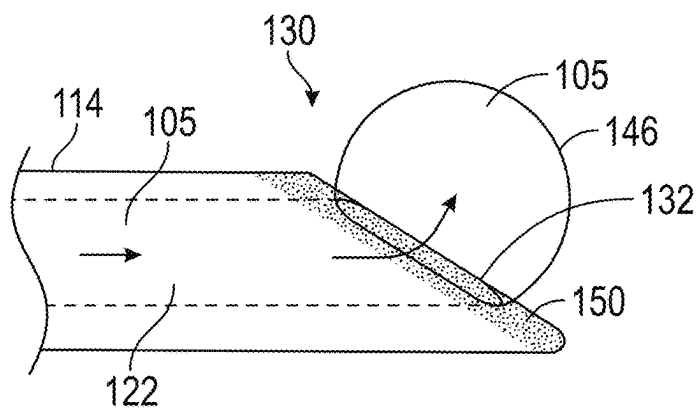


FIG. 4A

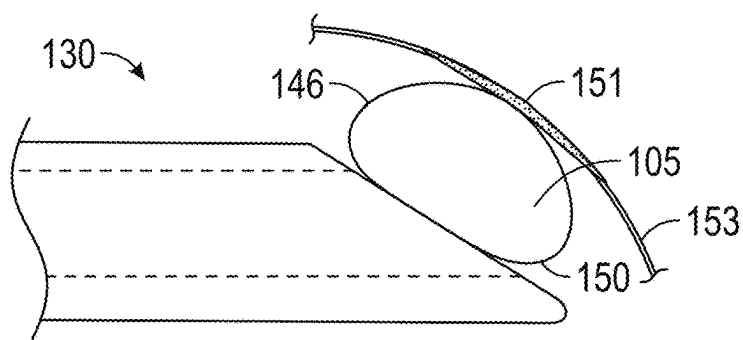


FIG. 4B

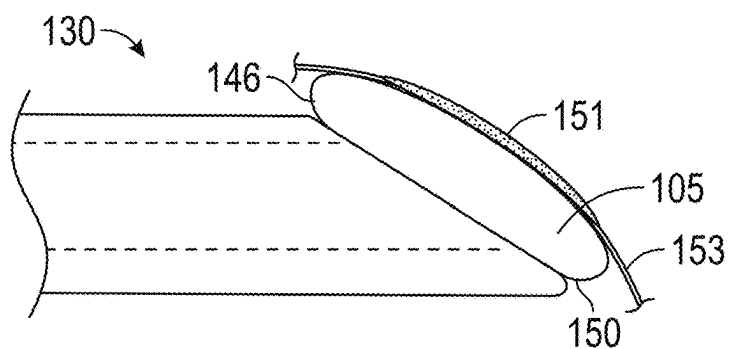


FIG. 4C

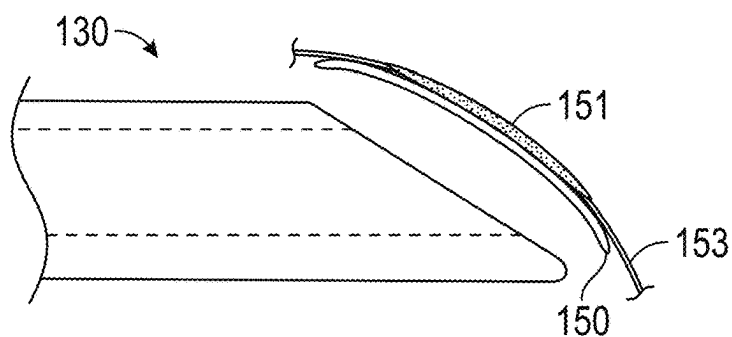


FIG. 4D

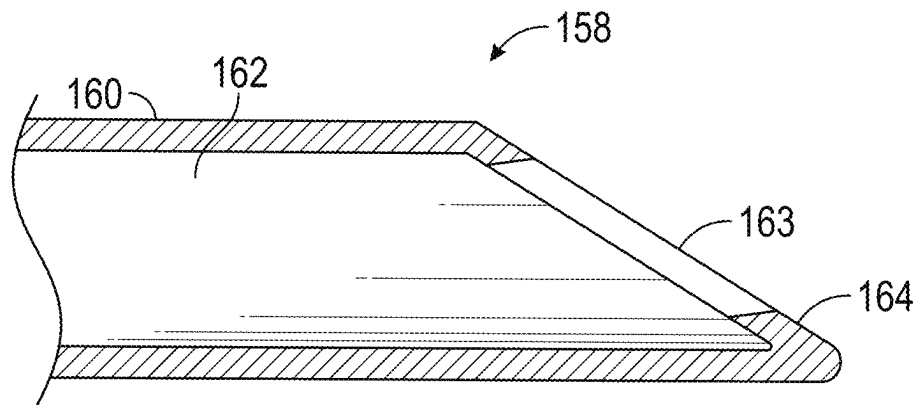


FIG. 5

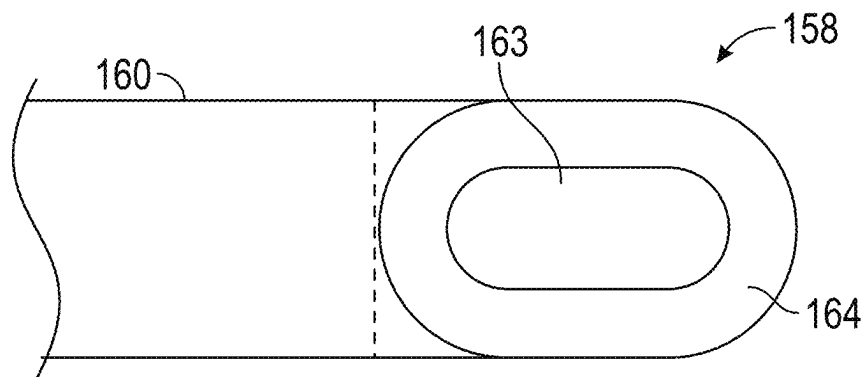


FIG. 6

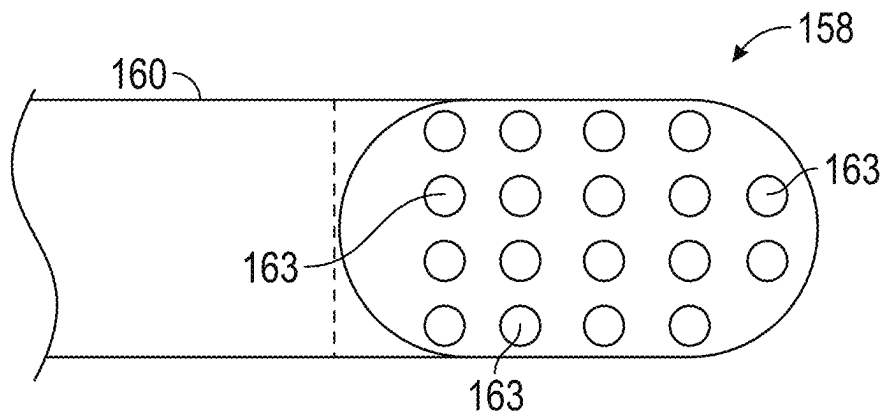


FIG. 7

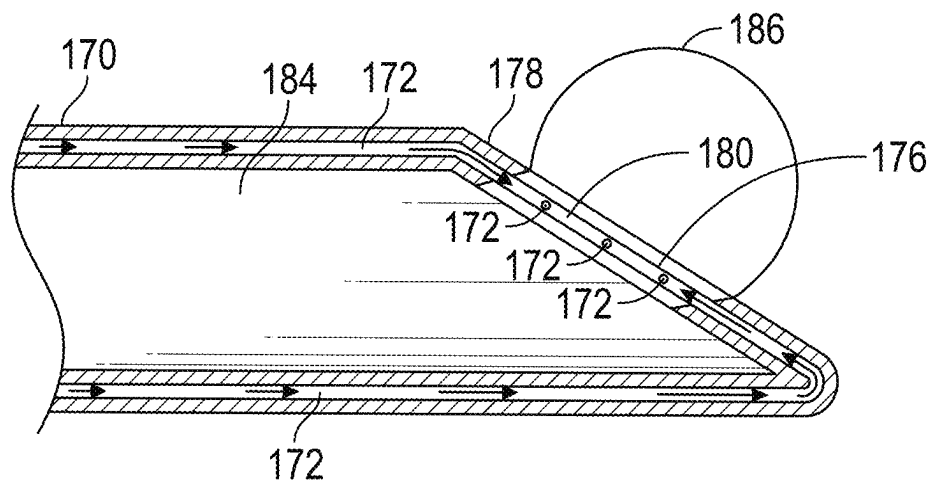


FIG. 8

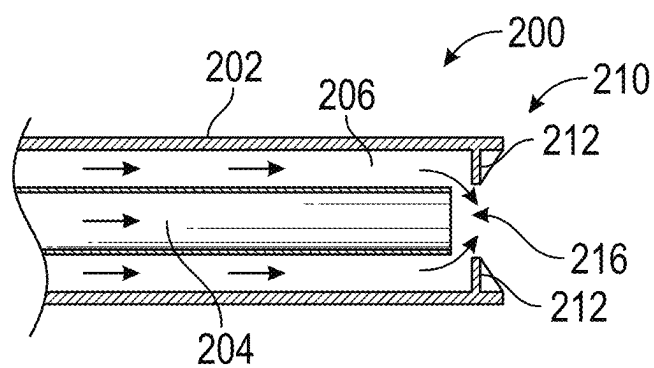


FIG. 9

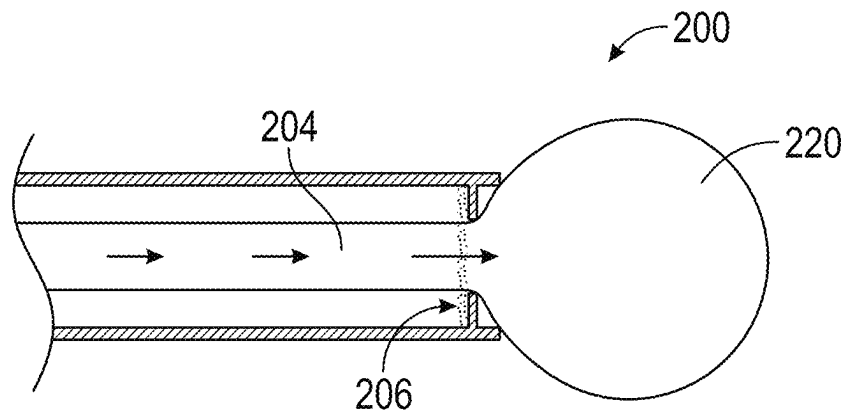


FIG. 10

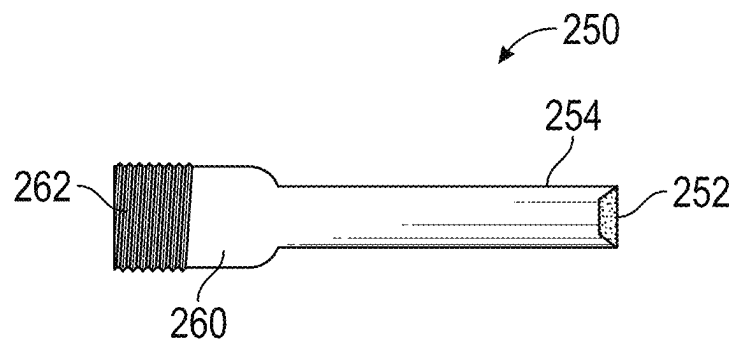


FIG. 11

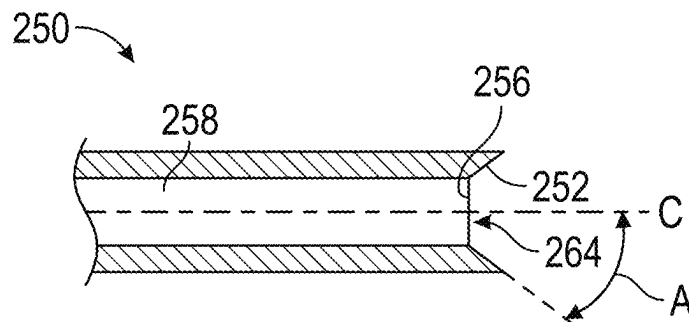


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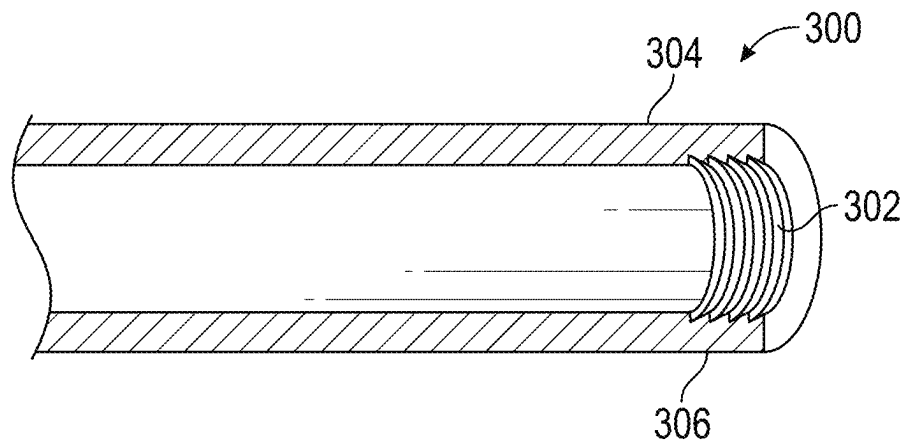


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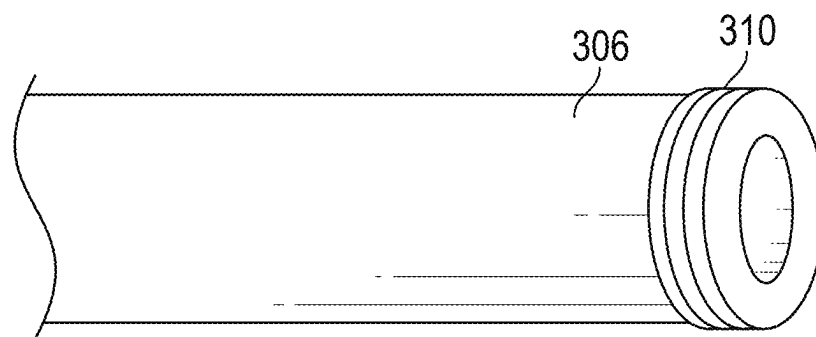


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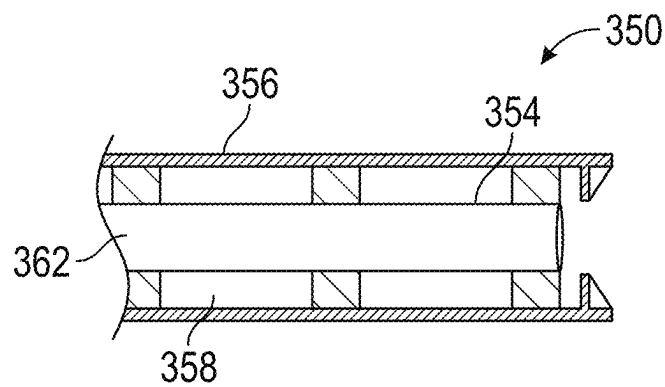


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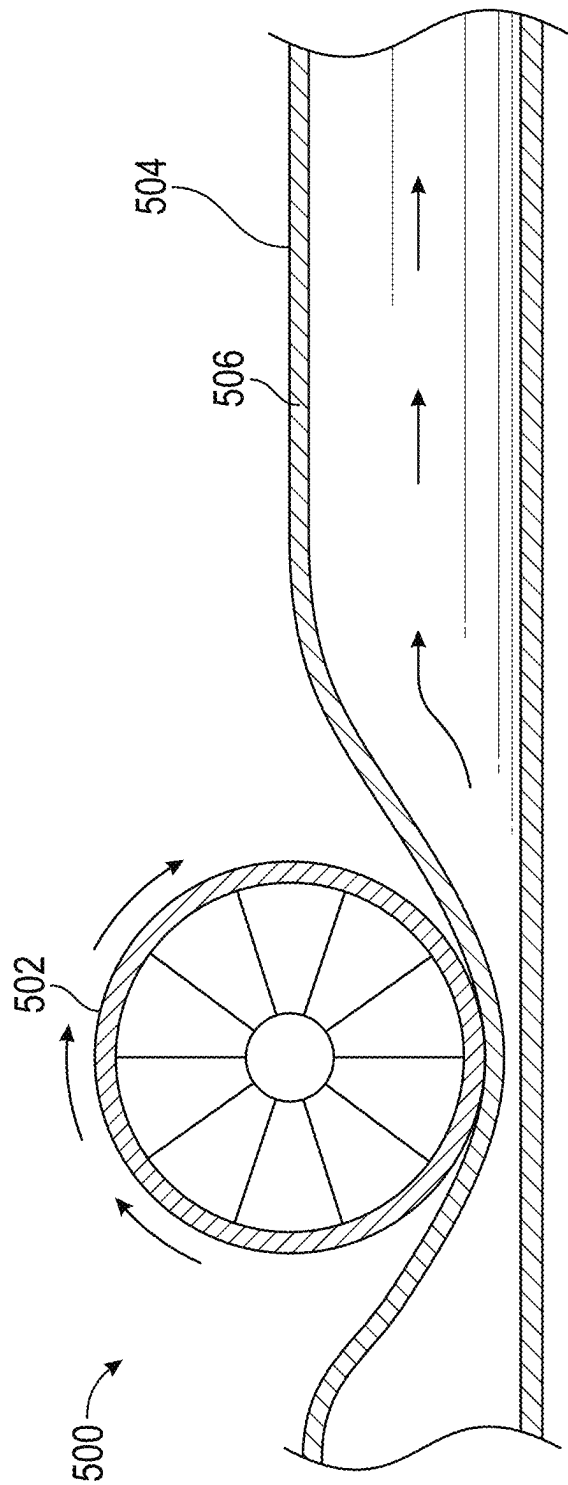


FIG. 16A

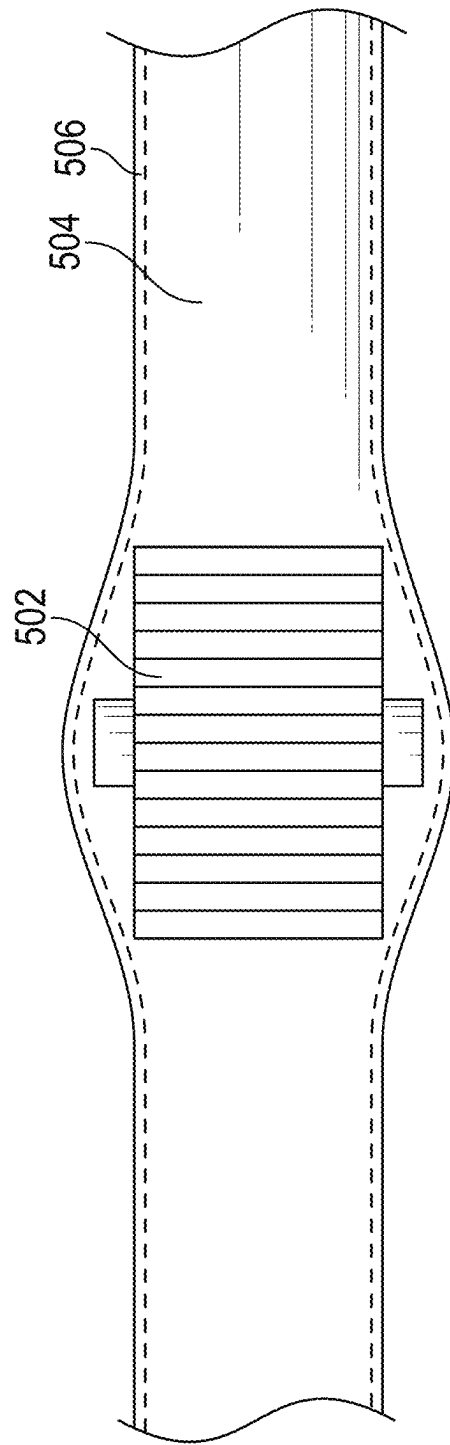


FIG. 16B

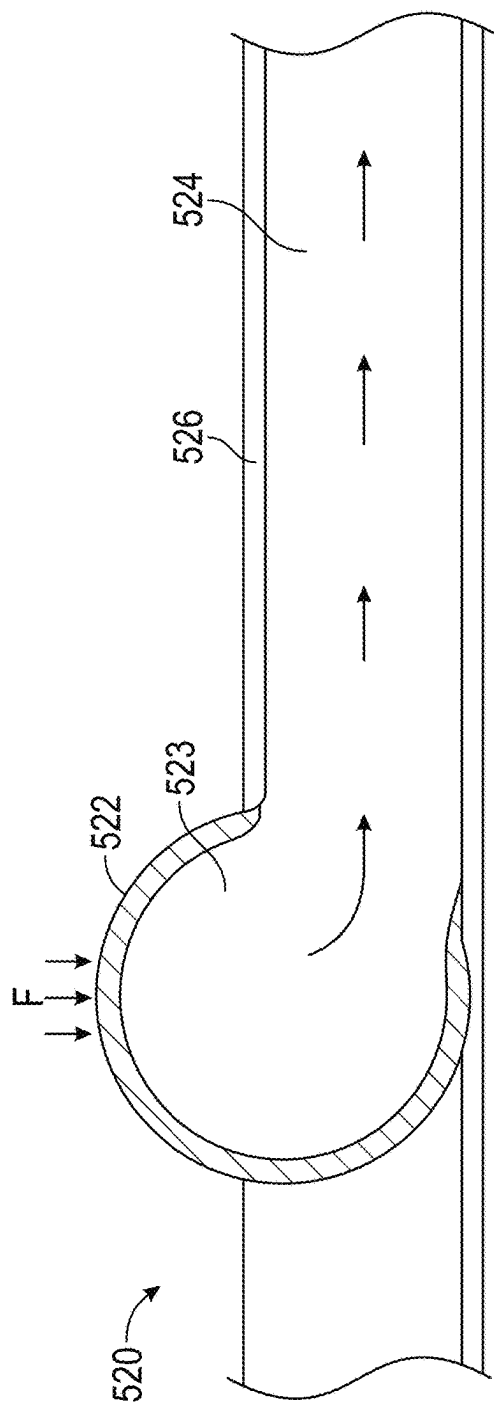


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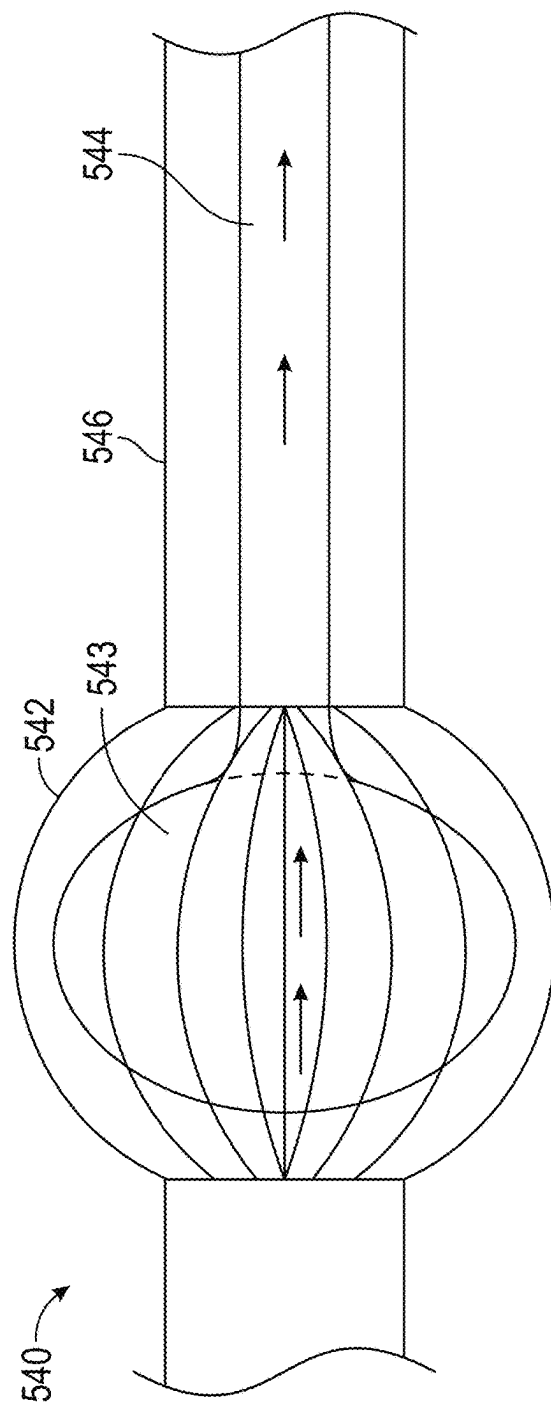


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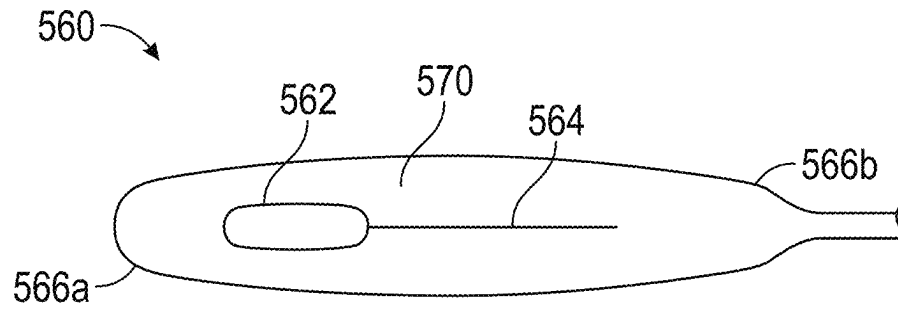


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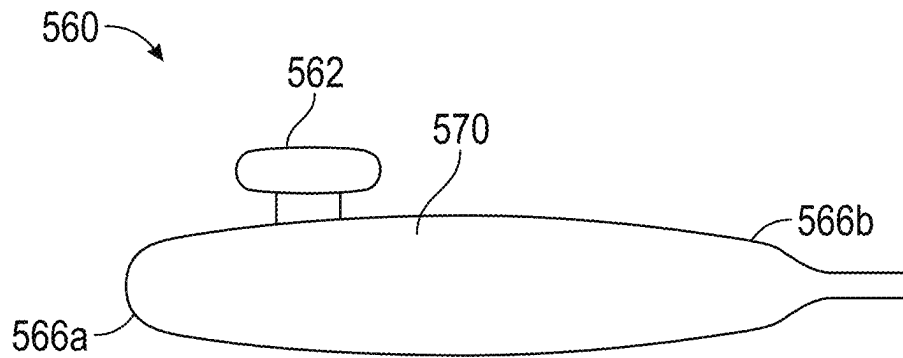


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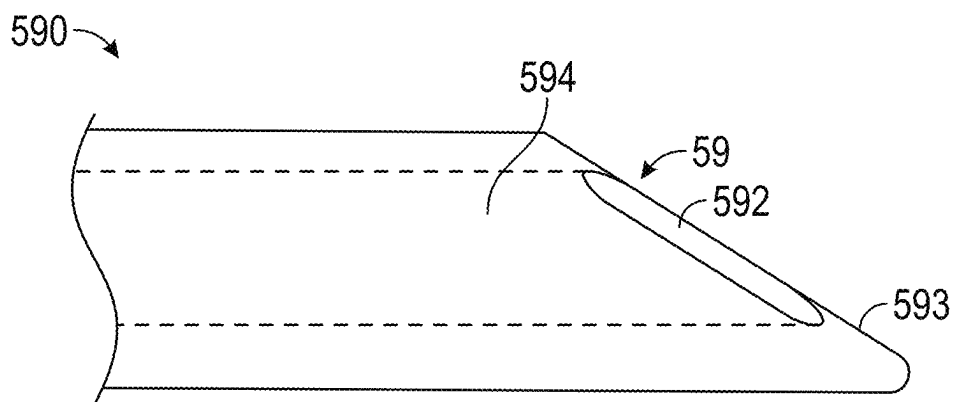


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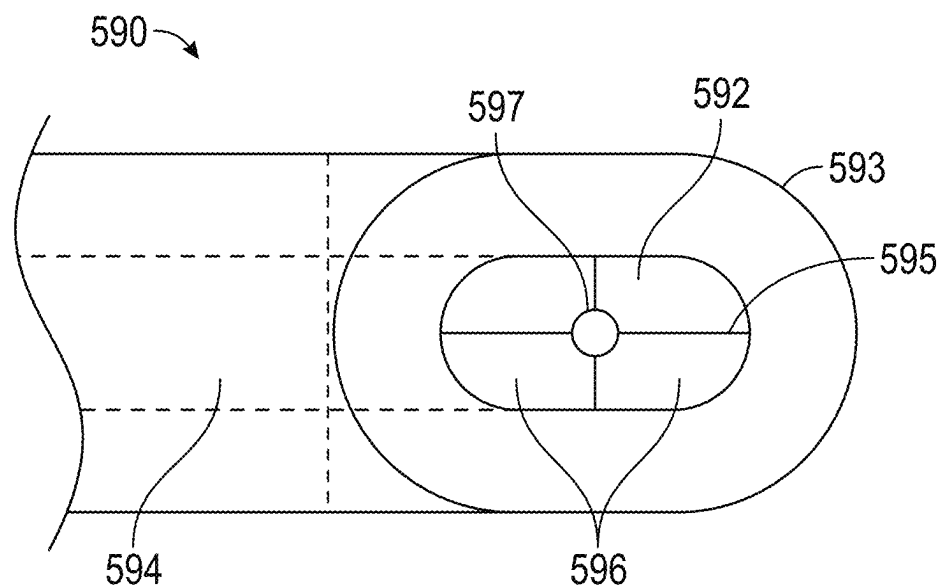


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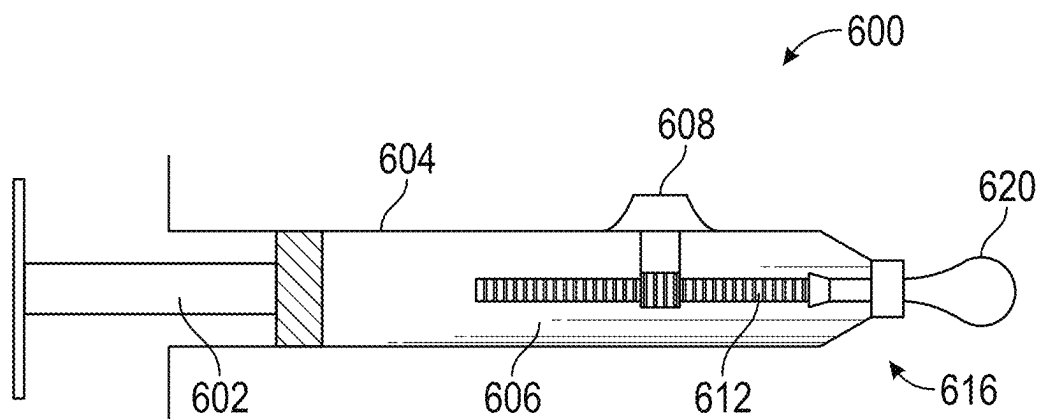


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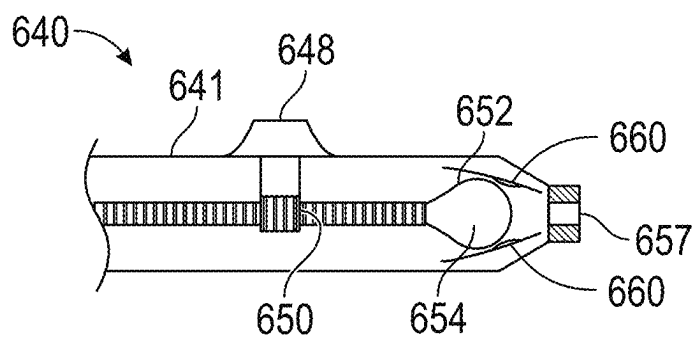


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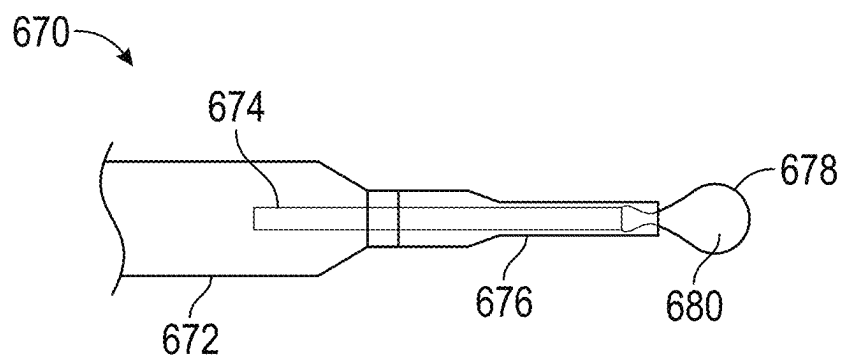


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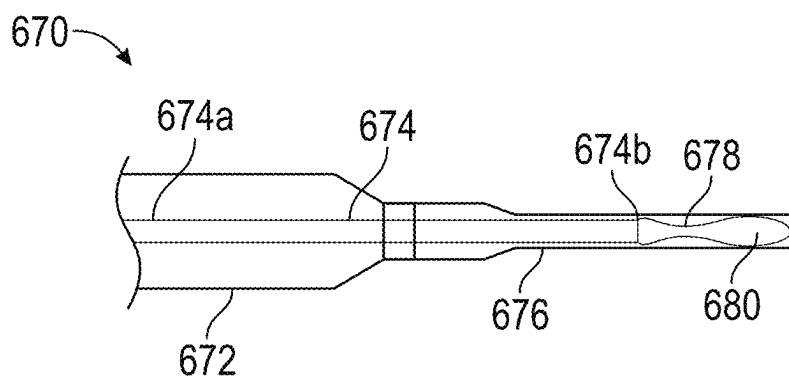


FIG. 25B

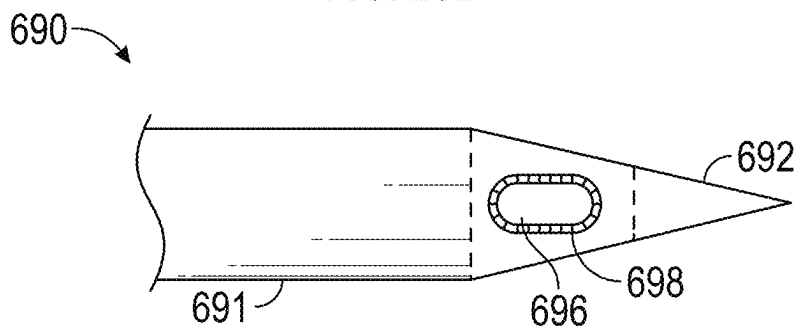


FIG. 26

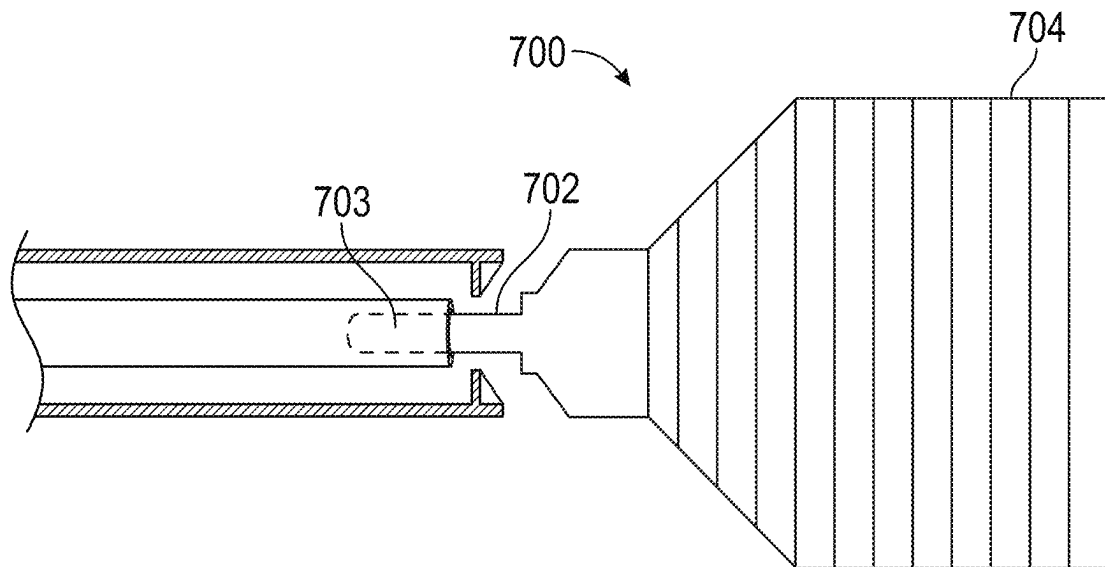


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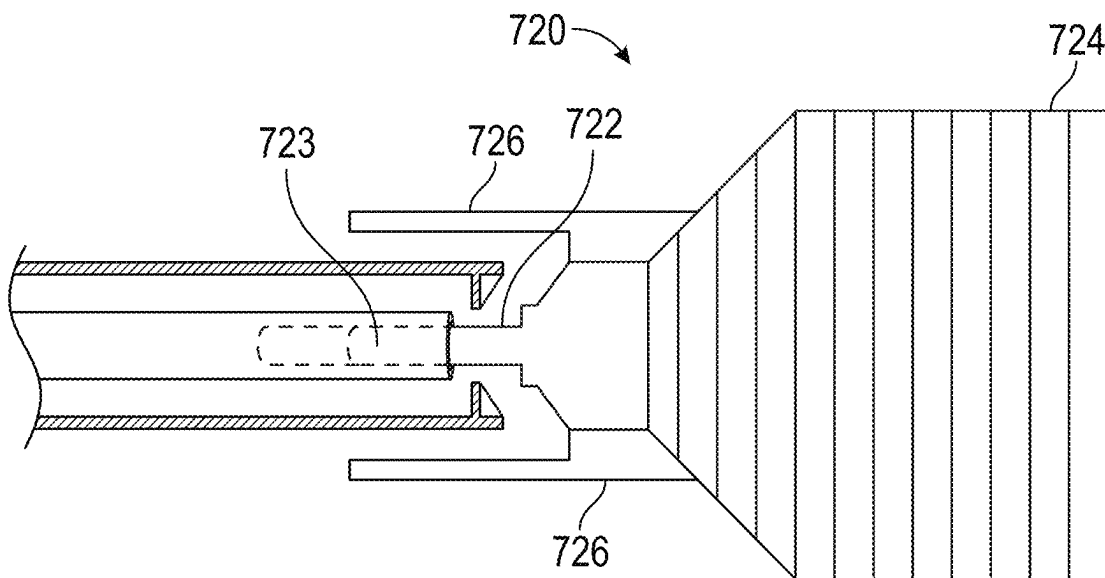


FIG. 28

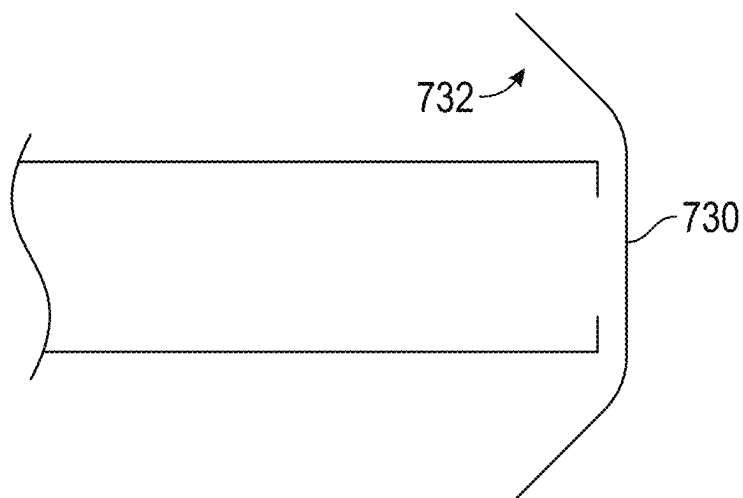


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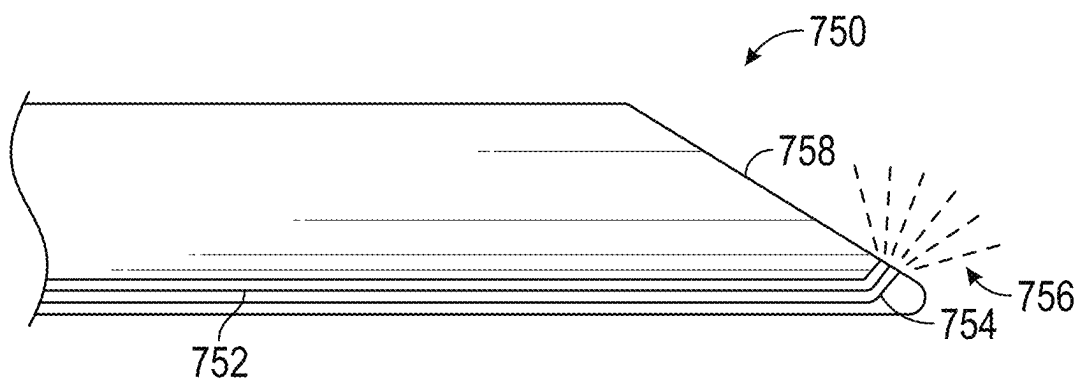


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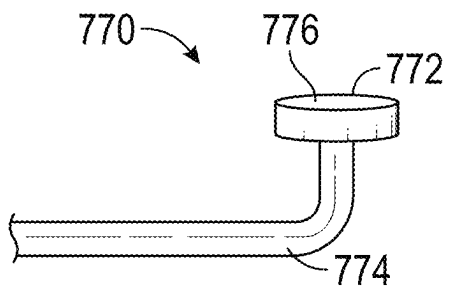


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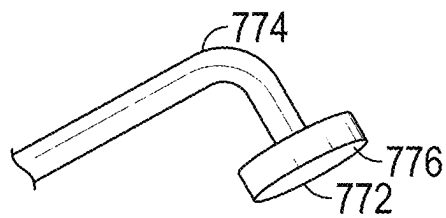


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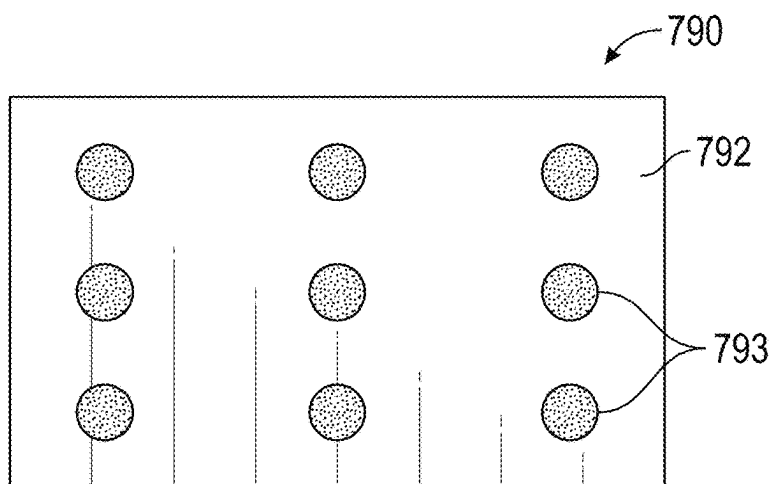


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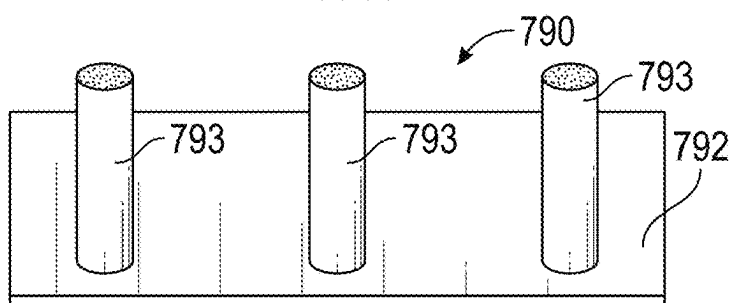


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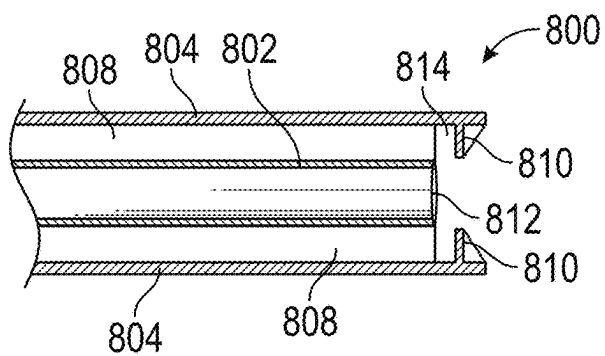


FIG. 35

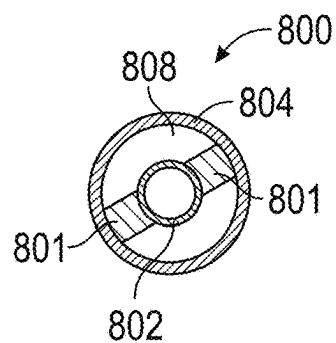


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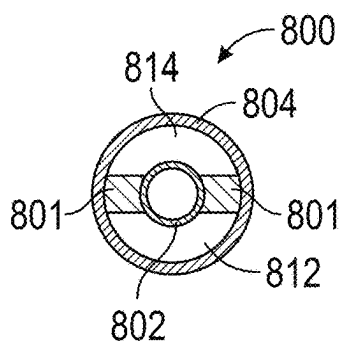


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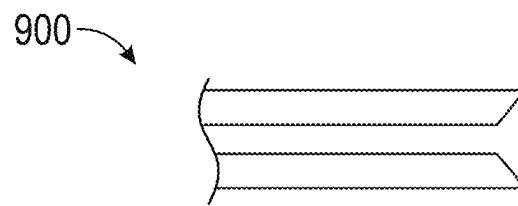


FIG. 38

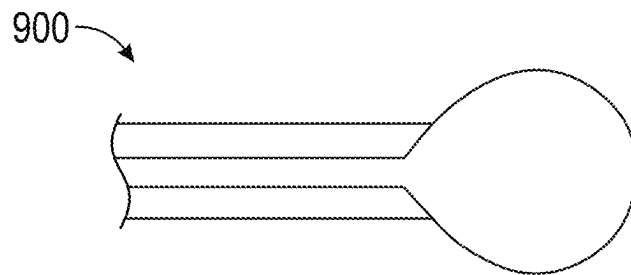


FIG. 39

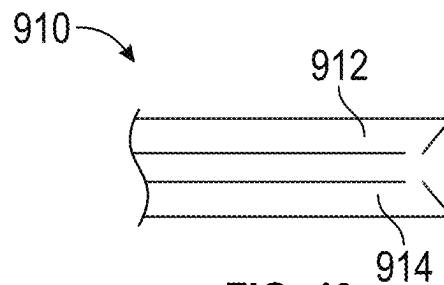


FIG. 40

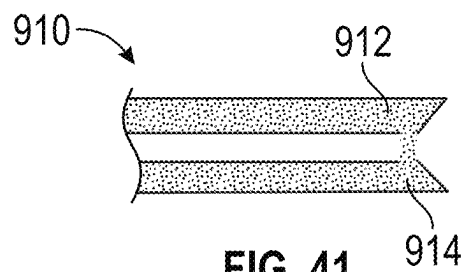


FIG. 41

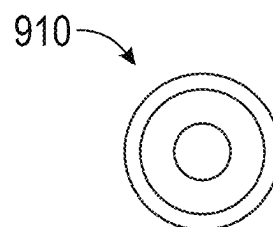


FIG. 42

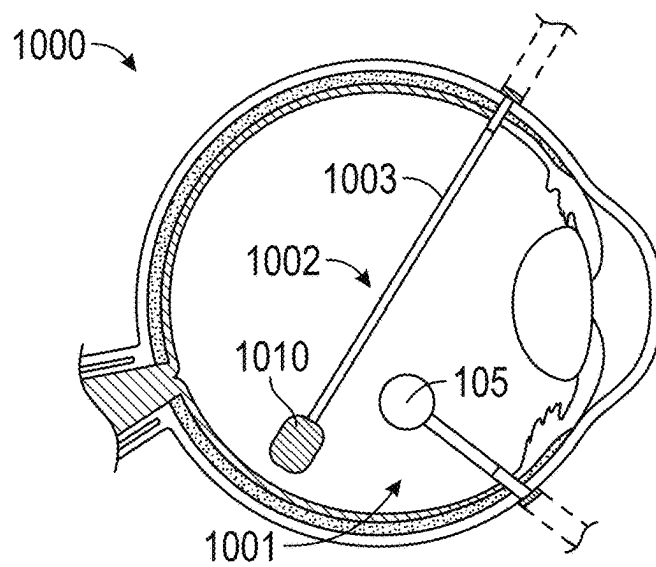


FIG. 43

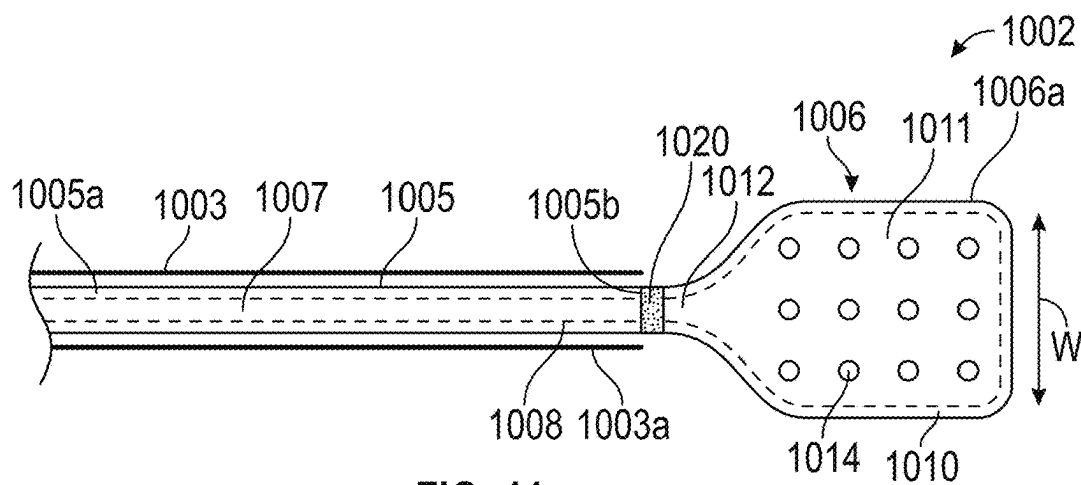


FIG. 44

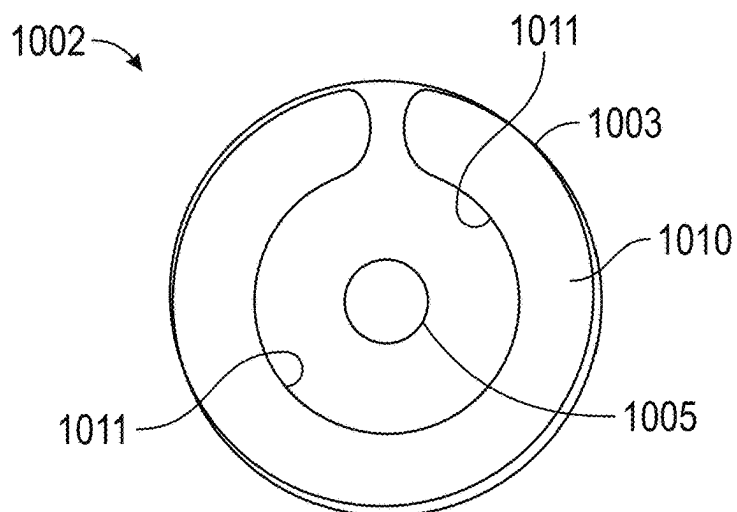


FIG. 45

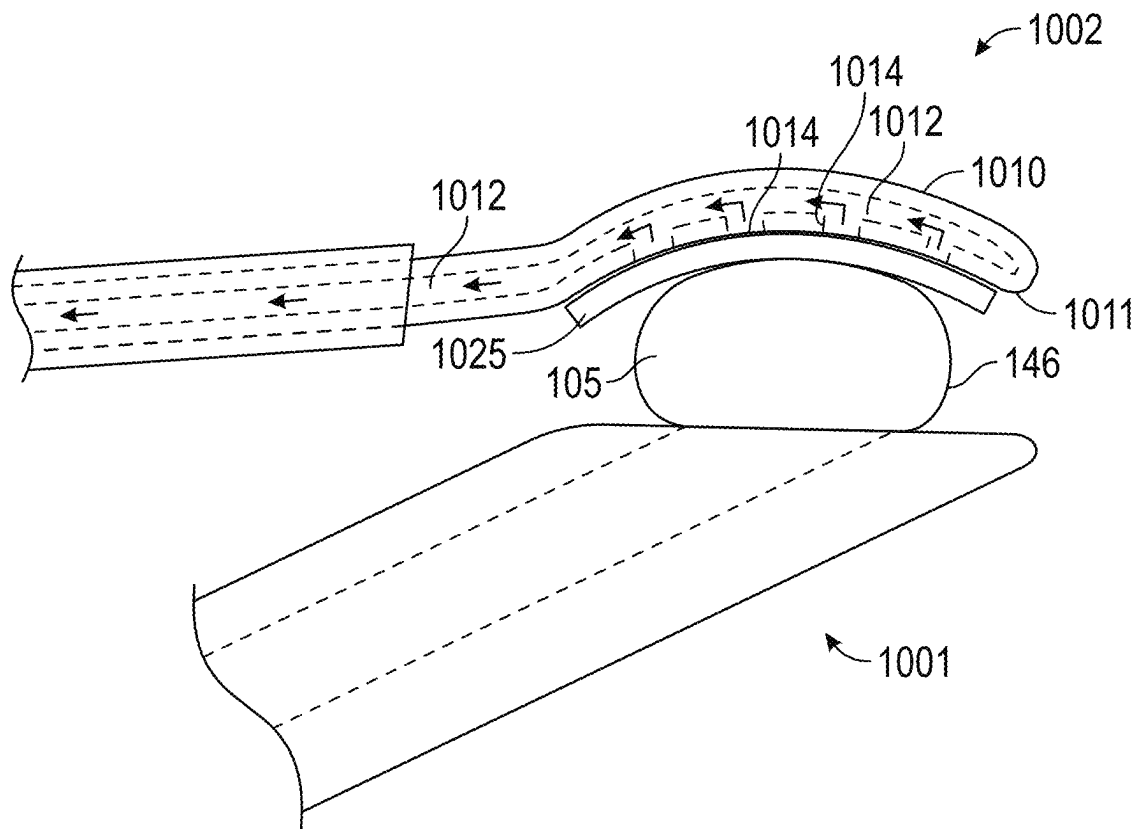


FIG. 46

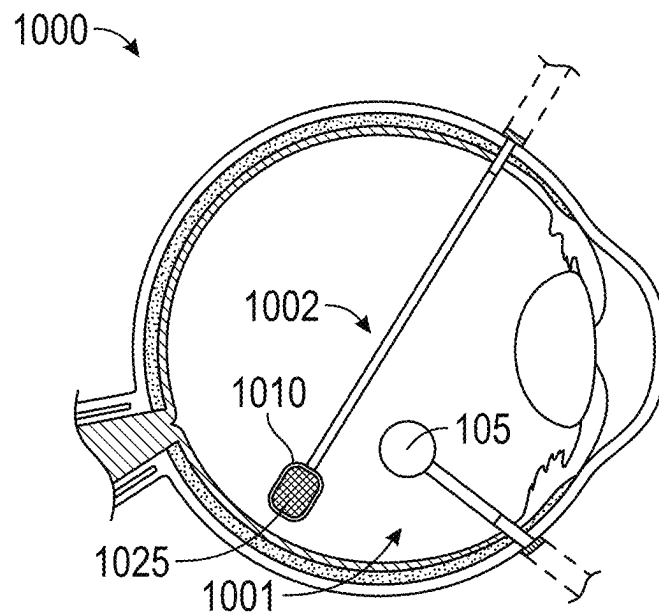


FIG. 47

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**DEVICES AND METHODS FOR TARGETED
DELIVERY OF A SUBSTANCE****CROSS-REFERENCE TO RELATED
APPLICATIONS**

This application claims the priority benefit under 35 U.S.C. § 119(e) to U.S. Provisional Patent Application Ser. No. 62/700,768 (filed Jul. 19, 2018) and U.S. Provisional Patent Application Ser. No. 62/796,338 (filed Jan. 24, 2019), the entirety of both of which are hereby expressly incorporated by reference as if fully set forth herein.

FIELD

Systems, devices, kits and methods provided herein relate to generation of a thin membrane for targeted delivery of a therapeutic substance, which membrane can be formed as a bubble.

BACKGROUND

Existing methods for fixing macular holes and retinal detachments, associated with retinal discontinuities such as tears and holes, include perfluorocarbon, laser, cryotherapy, gas bubble injection, and the use of silicone oil. The use of perfluorocarbon for displacement of fluid out of the subretinal space may result in inadvertent migration of perfluorocarbon liquid into the subretinal space which can result in vision loss if it is located in the central macula. Incomplete removal of perfluorocarbon liquid from the vitreous cavity may result in associated visual phenomena that patients may find annoying. Use of laser and cryotherapy results in permanent retinal scar tissue formation and may contribute to epiretinal membrane and epiretinal scar tissue formation. Use of gas bubble injection results in limitations on activities of the patient and can cause complications such as elevated intraocular pressure and cataracts. Likewise, silicone oil can be problematic because it sometimes causes complications such as elevation of intraocular pressure, requires a second surgery for removal, and frequently leaves behind residual oil bubbles after attempted removal that patients may find visually distracting.

Existing methods for fixing leaking eye wall discontinuities such as incisions, sclerotomies, and lacerations include sutures and commercially available bioadhesives. Sometimes sutured discontinuities continue to leak. Sutures cause ocular irritation, pain, tearing, and sometimes local tissue reactions that included edema and inflammation of the episclera and conjunctiva.

One difficulty with some bioadhesives is that they may induce toxicity. Another difficulty is that bioadhesive can be difficult to apply to non-dependent ocular surfaces, both inside and outside the eye, due to the effect of gravity on the bioadhesive as it leaves the tip of the bioadhesive delivery device and other factors. For example, in an eye with a retinal tear that is located on the lateral or superior aspect of the eye, gravity may make it difficult to apply the bioadhesive to the retinal tear since gravity may cause dripping or drooping of the bioadhesive as soon as the adhesive is expelled from the applicator device, preventing the accurate application of the adhesive over the retinal discontinuity. The same problem occurs when attempting to apply liquid or viscous materials to lateral and inferior outer eye surfaces. Another difficulty with bioadhesives is that they can be difficult to apply to large discontinuities of retina and other ocular surfaces, as well as tissue discontinuities that have an

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irregularly shaped margin or margins that are at different elevations from one another. Challenges in applying bioadhesives to retinal discontinuities can be even more difficult if attempted using a transcleral, subretinal approach.

SUMMARY

Arrangements disclosed herein include devices, systems, and methods for applying a substance such as, optionally, a bioadhesive, to a surface of an object. The tissue can be a biological or a non-biological (e.g., non-living) tissue and can include medical scaffolding, patches, covers, grafts, or other objects used in medical and non-medical applications. The object can be a tissue such that the devices and methods disclosed herein apply the substance to the tissue surface. The tissue can be a damaged tissue of any of the types or applications disclosed herein, including without limitation retinal tissue or other ocular tissue. As used herein, any features, components or other details that are described relative to a component, a device, a system, method, or any arrangements thereof are meant to apply to any of the other similar or suitable components, devices, systems, methods, and arrangements thereof disclosed herein.

As will be described, the devices, systems, and methods disclosed herein are configured to apply a substance, which can optionally be a bioadhesive, to the target tissue as a bubble, thin membrane, or other similar shape or arrangement. Some nonlimiting examples of applications of any of the devices and methods disclosed herein include applying a bio-substance such as a bioadhesive to an ocular tissue such as for a retinal tear, retinal hole, retinal detachment, etc., applying bioadhesive to the sclera, conjunctiva, and cornea, applying a bio-substance such as a bioadhesive to gastrointestinal tissue, colon tissue, etc. for sealing perforations or tears therein, air filled ocular cavities or spaces, or otherwise. However, the arrangements of the devices and methods disclosed herein are not limited to the delivery of therapeutic substances or ocular, medical, or biological applications. Any of the devices and methods disclosed herein can be used for the delivery of any desired or suitable substance or material in any desired application—biological, non-biological, mechanical, or otherwise.

Any arrangements of the device for applying a substance to a tissue can optionally include a handle portion, an elongate body comprising a proximal end, a distal end, and an intermediate region extending therebetween, an exit port at a distal end, at least one fluid passageway (also referred to herein as a first passageway) in fluid communication with the exit port. The fluid passageway can extend from the handle toward the intermediate region or beyond to the distal end. The elongate body can comprise device can also include a tubing member or cannula which can have a proximal portion, a distal tip, and a body portion extending therebetween.

In any arrangements, the system can comprise, can be comprised of, or consist of a unitary device wherein all of the components of the system are connected (e.g., fluidically coupled) to one another. Additionally, in any arrangements, the system can comprise, can be comprised of, or consists of a non-unitary device wherein one or more components of the system can be detached or detachable from another component of the system or be provided completely separately from other components. Some arrangements of the device can modular, interchangeable components.

A variety of materials can be used to make the system and/or device. The handpiece or cannula, or a component

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thereof, can comprise plastic, metal, polyvinyl chloride, glass, acrylic, carbon fibers and/or wood.

The distal tip and/or the end surface of the distal tip in any arrangements disclosed herein can have a retention ridge, lip or rim comprising a flat surface configured to hold and maintain the bubble substance. The end surface of any devices disclosed herein can have an inward member circumferentially facing inward toward a center of the distal tip, or circumferentially extending into the cannula exit port. The inward member can be configured to retain the bioadhesive substance and aid in formation of the bioadhesive bubble or of the spherical film of bioadhesive substance. The distal tip can have multiple end surfaces. In some arrangements, the distal tip can have no more than one end surface. The distal tip can be blunt or rounded, or can have a blunt or a rounded end.

The distal tip of any arrangements can optionally have a circumferential edge, and four or more, or from two to eight, or from four to six flaps extending inwardly across the cannula exit port. One or more of the flaps can have a beveled edge. A variety of materials can be used to produce the flaps, including without limitation rubber, silicone, plastic, or an elastic substance.

Some arrangements of the system further comprise a substance supply channel within the device or that can be advanced into the device to aid in the production of bioadhesive a bubble. The substance supply channel may be within or exterior to the expansion fluid passageway of the device (sometimes referred to herein as a hand-piece) and/or the fluid passageway of the cannula. The substance supply channel can have an elongate body comprising a proximal end and a distal end comprising a substance supply tip and/or exit port. The substance supply channel can be internal to or surrounded by at least a portion of the expansion fluid passageway of the cannula body, and/or a fluid passageway of the cannula base or tip. The or cannula can have an outer wall and an inner wall, the inner wall which can optionally provide a boundary to a substance supply channel, and a space between the inner wall and the outer wall comprising the expansion fluid passageway of the device, the fluid passageway of the cannula body, or a fluid passageway of the cannula base or tip. The substance supply tip can have a surface that is beveled as compared to a longitudinal axis of the substance supply channel body and the substance supply exit port. The end surface of the substance supply tip can have a 1-15°, 15-30°, 30-45°, 45-60°, 60-75°, 75-89°, 91-105°, 105-120°, 120-135°, 135-150°, 150-165° or 165-179° angle, or an angle in a range comprising more than one of the aforementioned angle ranges, compared to the longitudinal axis of the substance supply channel body. The substance supply channel can have a substance supply tip that can have one, from one to ten or more or from four to eight, or any number therebetween, or more substance supply tip exit ports. The substance supply channel can have a proximal end that extends into the fluid passageway of the cannula, and wherein the gas flows around the substance supply channel when the gas is blown through the fluid passageway of the cannula.

Some arrangements of the system further comprise circumferential or non-circumferential intermittent support structures connecting the outer wall and the inner wall, for example two or more from, from four to eight, or from six to ten or more, or any number therebetween, or more circumferential and/or non-circumferential intermittent support structures. These support structures can be used to provide a connection between an inner tube, chamber or wall and an outer tube, chamber or wall of the system.

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The substance supply channel can surround at least a portion of the expansion fluid passageway of the device, the fluid passageway of the cannula body, or a fluid passageway of the cannula base or tip. The cannula can have an outer wall and an inner wall, and a space between the inner wall and the outer wall, the space between the inner wall and the outer wall which can comprise the substance supply channel, and the inner wall comprising the expansion fluid passageway of the handpiece, the fluid passageway of the cannula body, or a fluid passageway of the cannula base or tip.

The substance supply channel can be a tube within the expansion fluid passageway of the device, the fluid passageway of the cannula body, and/or a fluid passageway of the cannula base or tip. The substance supply channel tube can extend within the cannula and be advanceable up to the distal tip of the device. The substance supply channel tube can have a distal end that reaches near the cannula exit port.

The substance supply channel, elongate body of the substance supply channel, substance supply tip and/or substance supply exit port can have any suitable size. The substance supply channel, elongate body of the substance supply channel, substance supply tip and/or substance supply exit port can have any suitable length.

Some arrangements of the system further comprise a second substance supply channel. For example, multiple internal supply chambers can be separated by support structures connecting an outer wall and an inner wall. The substance supply channel and the second substance supply channel each can be configured to provide a separate substance, for example a polyethylene glycol solution and a trilycine amine solution. The substance supply channel and the second substance supply channel each can have separate substances; for example one chamber may comprise one bioadhesive substance, and the other chamber may comprise an activator substance. Other substances may also be provided for in the one or more chambers. For example, in one arrangement, a first chamber may contain the bioadhesive, a second chamber a curing agent, and a third chamber a medicament (e.g., anti-inflammatory, antibiotic, etc.) it is contemplated that the struts or support structures can separate two or more chambers. For example a series of non-continuous struts can provide support by connecting an outer wall to an inner wall, or by connecting an outer wall of distal tip to an outer wall of a fluid passageway or substance supply channel within the distal tip.

Any arrangements of the system can have a substance dispenser. The substance dispenser can have base with a well or a series of wells. The base can be made of plastic, metal or glass. The well or the series of wells can each hold a substance such as the bioadhesive substance.

Any suitable substances may be used. The bioadhesive substance can optionally include ReSure sealant, a polyethylene glycol hydrogel, a polymeric gel, a double layer hydrogel, cyanoacrylate, fibrin glue, a polyethylene glycol solution, or a trilycine amine solution. The bioadhesive substance can include a resin. In any arrangements, the substance bubble or the approximately spherical film of substance enclosing the expansion fluid can have any suitable diameter or size. The diameter can change, expand, and/or decrease within, between, or among any of the aforementioned ranges.

Some arrangements relate to a handheld device for applying a bioadhesive bubble to a retina, comprising: an elongate body comprising a first end and a second end, the first end comprising a handle, the second end comprising a cannula comprising an exit port, at least one expansion fluid pas-

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sageway extending from the first end towards the second end; a distal tip at the second end, and configured to generate a bioadhesive bubble from a bioadhesive when a gas or liquid flows through the expansion fluid passageway, maintain the bubble on the distal tip when the gas stops flowing through the expansion fluid passageway, and release the bubble when the bubble is brushed against a retina or when the gas flows again through the expansion fluid passageway, or when it can be released by heating or cooling the tip of the cannula or using a second instrument to disengage bubble from distal tip.

Some arrangements relate to a handheld device for applying a bioadhesive bubble to a retina, configured to generate a bioadhesive bubble from a bioadhesive when a gas or liquid flows through the internal fluid passageway, maintain the bubble on the distal tip when the gas stops flowing through the internal fluid passageway, and release the bubble when the bubble is brushed against a retina or when the gas flows again through the internal fluid passageway, or when heating or cooling the tip of the cannula or using a second instrument to disengage the bubble from the distal tip.

Some arrangements relate to a kit comprising a system or device described herein. Some arrangements of the kit further comprise a second system or device, and wherein the systems or devices are each disposable after a single or limited number of uses. Some arrangements of the kit further comprise an adhesive biomaterial, a substance dispenser, and/or a tip plug or protective cover.

Some arrangements relate to a use of a system, device or kit for applying the bioadhesive bubble to an eye or retina such as for repair of the eye or retina. Some arrangements relate to a method of repairing a retinal tear, comprising: generating, with a system, device or kit, a bioadhesive bubble comprising an approximately spherical film of bioadhesive substance enclosing an expansion fluid, and applying the bioadhesive bubble to the eye or retina from the system, device or kit.

Some arrangements relate to a system for applying a bioadhesive bubble to a retina, or other biologic tissue. The system can have a handpiece and/or a cannula. The handpiece can have an elongate body comprising a proximal end, a distal end, and an intermediate region extending therebetween, wherein at least a proximal portion of the intermediate region can have a handle, the distal end comprising an exit port, and/or at least one expansion fluid passageway in fluid communication with the exit port and extending proximally towards the intermediate region. The cannula can include a cannula base, a distal tip, and/or a cannula body extending therebetween. The cannula body can include a fluid passageway configured to fluidically communicate with the exit port and receive an expansion fluid passed through the expansion fluid passage and exit port the distal tip can have a flat, non-beveled surface comprising a cannula exit port at the distal end of the distal tip, the surface configured to support a bioadhesive bubble generated when expansion fluid flows through the expansion fluid passageway to the at cannula exit port at the distal end of the distal tip and through a bioadhesive substance contained within the handpiece or cannula, thereby generating an approximately spherical film of bioadhesive substance enclosing the expansion fluid. There can be no bevel at the distal tip or within the cannula exit port.

In a first aspect, a device for applying a bubble of a substance to a tissue surface, the device comprising a cannula having a proximal end portion, a distal end portion, and an intermediate portion extending therebetween, a distal tip at the distal end portion of the cannula, the distal tip

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having a bubble support surface and an exit port extending through the bubble support surface, an expansion fluid passageway extending through at least the intermediate portion and the distal portion of the cannula, the expansion fluid passageway being in fluid communication with the exit port, a source of an expansion fluid, and an actuator coupled with the source of the expansion fluid and configured to selectively advance the expansion fluid through the expansion fluid passageway and the exit port upon actuation of the actuator. In an operable state, the distal tip can be configured to support a layer of the substance on the bubble support surface so that the layer of the substance completely covers the distal port. Additionally, when the device is in the operable state, the device can be configured such that the advancement of the expansion fluid from the fluid source through the exit port causes at least one bubble of the substance to form from the layer of the substance on the bubble support surface of the distal tip. Additionally, the device can be configured such that at least a portion of the bubble can be transferred from the distal tip to the tissue surface so as to treat a defect on the tissue surface. This can be achieved, in any arrangements disclosed herein, by moving the cannula, the distal tip, applicator portion, the loop, or device or component so that the bubble, the substance, the loop, the applicator portion, or otherwise over the tissue surface and/or the defect in the tissue surface so that the substance is spread about the tissue surface and/or defect to the desired amount, which can be performed in multiple steps.

The device for applying a bubble of a substance to a tissue surface can optionally include one or more of the following features, in any combination: (a) wherein the substance can be a bioadhesive; (b) wherein the bubble surface comprises a retention ridge, lip and/or rim configured to support the bubble; (c) having a concave curved depression formed in the bubble support surface; (d) wherein the device can be configured to support a plurality of bubbles on the bubble support surface; (e) having a handle portion at a proximal end of the device, the handle portion configured to support at least the actuator and the source of expansion fluid; (f) wherein the device can be configured to form a bubble that comprises a spherically shaped film of the substance at least partially enclosing the expansion fluid advanced through the exit port; (g) wherein the device can be configured to form a bubble that comprises a spherically shaped film of the substance that only partially encloses the expansion fluid advanced through the exit port; (h) wherein the actuator comprises a compressible bladder configured to expel the expansion fluid from the source of the expansion fluid within the bladder through the expansion fluid passageway and the exit port; (i) wherein the expansion fluid actuator comprises a roller wheel moveable along a compressible bladder configured to expel the expansion fluid from the source of the expansion fluid within the bladder through the expansion fluid passageway and the exit port; (j) wherein the expansion fluid actuator comprises a syringe; (k) further having a substance supply channel configured to supply the substance to the distal tip of the device; (l) wherein the substance supply channel can be in fluid communication with a substance supply source; (m) wherein the substance supply channel can be integrated within the cannula; (n) wherein the substance supply channel comprises an elongate body having a proximal end and a distal end having distal tip with at least one opening therein, the elongate body being advanceable through the expansion fluid passageway to the distal tip at the distal end portion of the cannula; (o) wherein the substance supply channel can be internal to or sur-

rounded by at least a portion of the expansion fluid passageway of the cannula; (p) wherein the cannula comprises an outer wall and an inner wall, the inner wall having the substance supply channel, and a space between the inner wall and the outer wall having the expansion fluid passageway of the device; (q) further having a second substance supply channel; (r) wherein the substance supply channel and the second substance supply channel each comprises a separate bioadhesive substance and/or activator; (s) wherein the bubble support surface can be beveled; (t) further having a substance dispenser; and (u) further having a cautery component configured to increase the temperature of at least the distal tip of the device.

In another aspect, a system for treating a defect on a tissue surface, having a first device for generating a bubble of a substance for treating a defect on the tissue surface, the first device having a first sleeve having a proximal end portion, a distal end portion, and an intermediate portion extending therebetween, a distal tip at the distal end portion of the first sleeve, the distal tip having an exit port extending through the distal tip, a fluid passageway extending through at least the intermediate portion and the distal portion of the first sleeve, the fluid passageway being in fluid communication with the exit port, and a supply of the substance.

The system for treating a defect on a tissue surface can optionally include one or more of the following features, in any combination: (a) wherein the device is configured to support a layer of the substance over the exit port in the distal tip when the first device is in an operable state; (b) wherein, when the first device is in the operable state, the first device can be configured such that passing a fluid through the exit port causes at least one bubble of the substance to form from the layer of the substance on the bubble support surface of the distal tip, the bubble having a spherically shaped film surface; (c) further having a second device for applying the bioadhesive substance to a retina, the second device having a second sleeve having a proximal end, a distal end, and a passageway extending along a length of the sleeve from the proximal end to the distal end of the sleeve, and an applicator having an elongate body and an applicator tip coupled with a distal end of the elongate body movable within the second sleeve, wherein the applicator portion can be self-expandable upon exit from the distal end of the second sleeve from a first collapsed state to a second expanded state, wherein the applicator tip has a larger width in the second expanded state; (d) wherein the system can be configured such that the at least one bubble can be transferred from the first device to the applicator tip of the second device and from the applicator tip of the second device to the tissue surface having the defect; and (e) further having a patch removably supported on a surface of the applicator tip, the patch being supported so that the at least one bubble can be transferred from the first device to the patch supported by the applicator tip of the second device and so that the patch can be transferred from the applicator tip of the second device to the tissue surface having the defect.

In another aspect, a method of repairing a defect in a retinal tissue, comprising advancing a substance supply device having a cannula and a distal tip toward the defect, providing a layer of a bioadhesive substance over an exit port in the distal tip of the device so that the layer of bioadhesive substance completely covers the exit port, forming a bubble of the bioadhesive substance on a support surface of the distal tip by advancing an expansion fluid through the exit port and the layer of bioadhesive substance, wherein the bubble can be at least partially attached to the support surface of the distal tip and has an approximately

spherically shaped film that extends away from the support surface of the distal tip, and transferring the bubble to the retinal tissue so as to at least partially cover the defect in the retinal tissue.

The method of repairing a defect in a retinal tissue can optionally include one or more of the following features, in any combination: (a) transferring a plurality of bubbles of the bioadhesive substance to the defect and/or the retinal tissue adjacent to the defect; and (b) transferring the bubble to the retinal tissue so as to at least partially cover the defect in the retinal tissue comprises transferring at least one bubble of the bioadhesive substance to a first surface of the patch and positioning the patch over the defect so that the first surface of the patch having the bioadhesive substance thereon can be in contact with at least the retinal tissue adjacent to the defect.

In another aspect, a handheld device for applying a bioadhesive bubble to a retina, having a proximal end having a handle, a distal end having a distal tip, and an internal air fluid passageway from the proximal end to the distal end. The handheld device for applying a bioadhesive bubble to a retina can optionally include one or more of the following features, in any combination: (a) wherein the device can be configured to generate a bioadhesive bubble from a bioadhesive when a gas flows through the internal air fluid passageway; (b) wherein the device can be configured to maintain the bubble on the distal tip when the gas stops flowing through the internal air fluid passageway; and (c) wherein the device can be configured to release the bubble to the tissue of the retina when at least the bubble can be advanced into contact contacted against a tissue of the retina.

In another aspect, a handheld device for applying a bioadhesive substance to a retina, having a tubular body having a first end and a second end, an exit port at the second end, and a first passageway extending from the first end to the exit port, an elongate body having a loop at a distal end thereof, the elongate body being advanceable and retractable within the first passageway so that the loop can be extended out of the exit port and into contact with a tissue surface of the retina, and a source of a bioadhesive substance configured to be applied to the loop, wherein the device can be configured such that the bioadhesive substance can be applied to the loop and the bioadhesive substance can be transferred from the loop to the tissue surface of the retina by contacting the tissue surface of the retina with the bioadhesive substance that is on the loop.

In another aspect, a use of a system, device, or method of any of the foregoing aspects.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a side view of a non-limiting example of a device for providing a substance.

FIG. 2 is a side view of a distal portion of the device illustrated in FIG. 1.

FIG. 3 is a top view of the distal portion of the device illustrated in FIG. 1.

FIG. 4A is a side view of an example of a distal tip and a bubble formed from a substance.

FIG. 4B shows the distal tip and bubble being advanced toward a desired treatment location.

FIG. 4C shows the distal tip and bubble being pressed into contact with a defect in the target tissue or surface in the desired treatment location.

FIG. 4D shows the distal tip being withdrawn after the bubble has ruptured or dissipated, depositing the substance over the defect in the desired treatment location.

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FIG. 5 is a section view of a distal portion of a non-limiting example of a substance supply channel.

FIG. 6 is a top view of the distal portion of the substance supply channel illustrated in FIG. 5.

FIG. 7 is a top view of another non-limiting example of a distal portion of a substance supply channel.

FIG. 8 is a section view of another non-limiting example of a device for providing a substance.

FIG. 9 is section view of another non-limiting example of a distal portion of a device for providing a substance.

FIG. 10 is section view of another non-limiting example of a distal portion of a device for providing a substance, showing a bubble formed from a substance.

FIG. 11 is a side view a non-limiting example of a cannula.

FIG. 12 is a section view of a non-limiting example of a distal tip.

FIG. 13 is a section view of a non-limiting example of an interior of a distal tip with grooves.

FIG. 14 is a side view a non-limiting example of an exterior of a distal tip with grooves.

FIG. 15 is a section view of another non-limiting example of a distal tip.

FIG. 16A is a section view a non-limiting example of a device for providing an expansion fluid, the device comprising a roller wheel and a compressible bladder.

FIG. 16B is a top view of the device for providing an expansion fluid shown in FIG. 16A.

FIG. 17 is a section view of a non-limiting example of a device for providing an expansion fluid comprising a compressible bladder.

FIG. 18 is a top view of another non-limiting example of a device for providing an expansion fluid comprising a compressible bladder.

FIG. 19 is a top view of a non-limiting example of a device for providing an expansion fluid comprising a moveable slider.

FIG. 20 is a side view of the device for providing an expansion fluid comprising a moveable slider.

FIG. 21 is a side view of a non-limiting example of a distal tip comprising flaps.

FIG. 22 is a top view of the distal tip shown in FIG. 21.

FIG. 23 is a section view of a non-limiting example of a device for providing a substance.

FIG. 24 is a section view of another non-limiting example of a device for providing a substance.

FIG. 25A is a section view of another non-limiting example of a device for providing a substance, showing the loop in an extended position.

FIG. 25B is a section view of the device shown in FIG. 25A, showing the loop in a withdrawn position within the cannula.

FIG. 26 is a top view of a non-limiting example of a system or device with a trans-scleral tip.

FIG. 27 is a section view of a distal portion of a non-limiting example of device for providing a substance.

FIG. 28 is a section view of a distal portion of another non-limiting example of device for providing a substance.

FIG. 29 is a side view of a distal portion of another non-limiting example of a tip cover.

FIG. 30 is a side view of a non-limiting example of a cannula with one or more fibers or wires for light or cautery.

FIG. 31 is a first side view of a distal portion of a non-limiting example of a cannula.

FIG. 32 is a second side view of the cannula shown in FIG. 31.

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FIG. 33 is a top view of a non-limiting example of an arrangement of a substance dispenser.

FIG. 34 is a front view of the arrangement of the substance dispenser shown in FIG. 33.

FIG. 35 is a section view of a distal portion of another non-limiting example of a cannula.

FIG. 36 is a section view of another non-limiting example of a cannula.

FIG. 37 is a section view of another non-limiting example of a cannula.

FIG. 38 is a side view of a non-limiting example of a distal tip.

FIG. 39 is a side view of a non-limiting example of a distal tip with a bubble.

FIG. 40 is a side view of a non-limiting example of a distal tip.

FIG. 41 is a side view of a non-limiting example of a distal tip with a substance within the distal tip.

FIG. 42 is an end view of a non-limiting example of a system or device as described herein.

FIG. 43 illustrates a non-limiting example of a system having an applicator device and a substance providing device.

FIG. 44 is a top view of a non-limiting example of an applicator device, showing an applicator tip of the device in an open state.

FIG. 45 is an end view of the applicator device shown in FIG. 44, showing the applicator tip of the device in a closed state.

FIG. 46 is a side view of a non-limiting example of an applicator device and a substance providing device, showing the substance providing device depositing a bubble of a substance against a surface of a patch supported by an applicator tip of the applicator device.

FIG. 47 is another illustration of the system having the applicator device and the substance providing device.

DETAILED DESCRIPTION

In the following detailed description, reference can be made to the accompanying drawings, which form a part hereof. In the drawings, similar symbols typically identify similar components, unless context dictates otherwise. The illustrative arrangements described in the detailed description, drawings, and claims are not meant to be limiting. Other arrangements may be utilized, and other changes may be made, without departing from the spirit or scope of the subject matter presented herein. It will be readily understood that the aspects of the present disclosure, as generally described herein, and illustrated in the Figures, can be arranged, substituted, combined, separated, and designed in a wide variety of different configurations, all of which are explicitly contemplated herein.

The arrangements disclosed herein comprise devices, systems, and methods for applying a substance to a surface, including tissue surfaces. In any arrangements disclosed herein, the substance can be a therapeutic substance which can be applied to a tissue. The tissue in any arrangements can be an ocular tissue. Examples of substances that can be used with any of the devices, systems, and methods disclosed herein include, without limitation, adhesives, bioadhesives, gels, hydrogels, thick liquids or semi-liquid treatment substances, double layer hydrogels, nonsolids and the like. As used herein, any use of the term substance is intended to include any of the types and examples of substances disclosed anywhere in this disclosure.

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Any arrangements of the systems, devices, and methods disclosed herein can be configured to generate thin membranes of the substance for delivery to a biological tissue, including without limitation, an ocular tissue such as for a retinal tear or detachment, or intraocular, intra-abdominal, intracranial, epidermal bioadhesive delivery. However, the arrangements of the devices, systems, and methods disclosed herein are not limited to the delivery of substances to ocular, medical, or biological applications. The devices, systems, and methods disclosed herein can be used for the delivery of any desired or suitable substance in any desired or suitable application, biological, non-biological, mechanical, or otherwise.

Some arrangements include creating a bubble out of an adhesive, for example, blowing a bubble. The bubble can include a thin film that can be applied to a surface such as a retina, that can be either attached or detached from the back of the eye. In any arrangements, the bubble can be applied to the surface of the retina in an air-filled eye, or applied to the surface of the retina in a fluid-filled eye. Benefits of applying the bioadhesive in this way include an improved ability of the adhesive to conform to and stick to the eye and retina, and decreased or improved drying, curing, or activation time of the adhesive.

Arrangements of the devices and methods disclosed herein are configured to permit the creation of a bubble or plurality of bubbles in a controlled fashion, so that the release of the bubble or bubbles from the delivery device can be controlled so as to be not released until a time that is desired. Additionally, the devices and methods disclosed herein optionally are configured to permit the reformation of a bubble, or the new formation of a bubble, after the film has been removed and dissipated by the formation of bubble or bubbles during the preceding operation of the device, to make it ready for the next operation. The devices disclosed herein can optionally be configured to be used in any orientation, including but not limited to horizontal, vertical, inverted, or otherwise so as to enable use with any desired tissue, including eye tissue, walls or roof of mouth, nose, vagina, or other tissue.

As used herein, the term bubble can mean a complete, spherical membrane or body of the substance, a spherically shaped or curved shape or film of the surface, including without limitation a half sphere shape of the substance, a single bubble or a plurality of bubbles, either separate or connected together such as in a foam. All uses of the term bubble herein are meant to include any one or all of these examples of a bubble set forth herein or elsewhere within the description.

For example and without limitation, any of the devices disclosed herein can be configured to form a bubble, which means that the device can be configured to form a spherical membrane or body of the substance, a spherically shaped or curved shape or film of the surface, including without limitation a partial sphere or half sphere shape of the substance, a complete sphere of the substance, a single bubble or a plurality of bubbles, either separate or connected together such as in a foam.

In any arrangements disclosed herein, the bubble or the approximately spherical film of bioadhesive substance can have a diameter ranging from approximately 0.1 mm to approximately 15 mm or more (up to at least 100 mm), or from approximately 0.5 mm to approximately 5 mm, or from approximately 1 mm to approximately 3 mm, or from and to any values within these ranges.

Some arrangements relate to a system and/or a device for generating a bioadhesive bubble and/or applying a bioad-

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hesive bubble to a surface. The surface can be a biological surface such as a retina. The system or device can optionally include an elongate body having a handle portion and a distal port. The elongate body can have a proximal end, a distal end, and an intermediate region extending therebetween. At least a proximal portion of the elongate body can include a handle portion. The distal end of the elongate body can have an exit port. In any arrangements, the elongate body, which can be a cannula as described in more detail below, can include at least one expansion fluid passageway or, optionally multiple (e.g., 2, 3, 4 or more) expansion fluid passageways extending partially or completely there-through. The expansion fluid passageway(s) can be in fluid communication with the exit port and extend proximally toward the intermediate region of the elongate body.

In any arrangements, the fluid passageway can be configured to enable a supply of a fluid, which can be a gas or other expansion media, through the body of the device toward the exit port. The fluid passageway can be in fluid communication with the exit port. In any arrangements, a fluid such as a gas or other expansion media can be advanced through fluid passageway to exert a positive pressure on the substance to cause the substance to expand or form a curved or spherically shaped film or one or more bubbles from the substance.

The expansion media can, for example, inflate or otherwise expand a film or membrane of the substance (optionally, a bioadhesive substance) prior to application of the bioadhesive to a target tissue or tissue region. In some arrangements, the surgeon can activate the source of the expansion media (which can be a gas) and provide the expansion fluid through the expansion passage. The expansion fluid can be passed through the device and out through the opening at the distal end of the device through a substance to produce a bubble and/or spread a film of the substance on a desired surface.

FIG. 1 illustrates a non-limiting example of an application device **100** that can be used to provide a substance (including, for example and without limitation, a bioadhesive) to a target tissue. The device **100** can include a handle **102** comprising an expansion fluid source **104**, which can optionally include an air bladder, that can be coupled with a body member **106** having a fluid passageway **107** extending therethrough. It shall be appreciated that other expansion fluid sources can be used, such as an air/gas supply line rather than an air bladder or any of the other types of expansion fluid sources disclosed herein or used in the industry, now or as later developed.

The expansion fluid source **104** can be filled with an expansion fluid **105** that can be used to generate the bubbles. The expansion fluid source **104** can be in fluid communication with the fluid passageway **107** so that the expansion fluid can be selectively released or communicated from the expansion fluid source **104** through the fluid passageway **107**. The body member **106** can optionally be formed from a flexible, rigid, semi-rigid or other suitable material or component.

In any arrangements disclosed herein, the expansion fluid can include a gas such as air, sulfur hexafluoride (SF₆), perfluoropropane (C₃F₈), and/or nitrogen or other inert gas, or any combination thereof. The expansion fluid can include a liquid such as water, a heavy liquid such as perfluoro-n-octane, a buffer, a solvent and/or an oil such as silicone oil. The expansion fluid enters and/or fills a bubble of bioadhesive substance, and/or causes the bubble to expand as the expansion fluid enters and/or fills the bubble. The expansion fluid can include 5-50%, 10-15%, 10-20%, 10-30%, 5-25%,

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5-15%, 15-25%, 15-20%, 20-25%, 20%, about 20%, 12%, about 12%, 14%, about 14%, 12-14%, or 11-15% SF₆. the expansion fluid can include 5-50%, 10-15%, 10-20%, 10-30%, 5-25%, 5-15%, 15-25%, 15-20%, 20-25%, 20%, about 20%, 12%, about 12%, 14%, about 14%, 12-14%, or 11-15% C₃F₈. For example, the expansion fluid may comprise 20% SF₆, 12% C₃F₈, and 68% air, or the expansion fluid may comprise 20% SF₆, 14% C₃F₈, and 66% air. The expansion fluid can also include a liquid such as water, a heavy solvent or liquid such as perfluoro-n-octane, an oil, or an oil-water mixture.

A connector **110** can be coupled with an end portion of body member **106** to enable the selective connection and disconnection of an end piece **114**, which can be or include cannula. The connector **110** can include a connector portion **116** that can, optionally be threaded that is configured to reversibly couple with a proximal end portion **118** of the cannula **114**. In any arrangements, the connector portion **116** can have an opening **120** or passageway therethrough that can be threaded and configured to threadedly engage with the proximal portion **118** of the cannula **114**. In some arrangements, the opening **120** can have internal threads configured to engage with external threads on the proximal portion **118** of the cannula **114**. Optionally, the connector **110** can have external threads thereon that are configured to engage internal threads on the inside of the proximal portion **118** of the cannula **114**.

The connector **110** can optionally comprise a Luer lock connector, configured to reversibly connect with the cannula **114**. The cannula **114** can have a generally uniform or consistent cross-sectional size along a length of the cannula **114**. In some arrangements, as illustrated, the proximal portion of the cannula **114** can have an increased cross-sectional size or diameter as compared to an intermediate and distal portion of the cannula **114**. The cannula **114** can have a passageway **122** therethrough that can be in fluid communication with the opening or passageway **120** in the connector **110** and the passageway **107** through the body member **106**. The cannula **114** can have an end portion **130** (also referred to herein as a distal tip) having an opening **132** therethrough that is in fluid communication with the passageway **122** that extends through the cannula **114**.

Additionally, the device **100** can optionally have one or more valves **140** such as a one-way valve or other flow restrictor at any location along a length of the passageway **107**. The valve **140** can prevent the escape or inadvertent discharge of the expansion fluid **105** within the passageway **107** or expansion fluid source **104** and/or can be configured to prevent backflow into the source **104**. One or more such valves can also be located in the cannula **114**.

Note that the components shown in FIG. 1, and in some other figures are not be drawn to scale. For example, in other arrangements, the cannula **114** can be longer or shorter than what is shown in FIG. 1, or can have a larger or smaller cross-sectional size as compared to the example illustrated in FIG. 1.

In any arrangements, the connector **110** can be configured to engage with any of a variety of different cannulas **114**. The variety of different cannulas **114** can have a range of different cross-sectional sizes and/or shapes, a variety of different lengths, and a variety of other different features and characteristics. With reference to the device **100** illustrated in FIG. 1, the distal tip **130** of the cannula **114** can be smaller or larger than as shown in FIG. 1 relative to one or more of the other components, including the connector **110** and the body member **106**.

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The cannula **114** can include a proximal portion or base **118**, a distal tip **130**, and a body **119** extending therebetween. As described, the cannula **114** can have an expansion fluid passageway **122** extending through the cannula body. The fluid passageway **122** can be configured to fluidically communicate with the exit port **130** and communicate an expansion fluid **105** through the expansion fluid passage **122** and exit port **132**.

As the expansion fluid flows through the substance located in or on the distal tip **130** of the cannula **114**, a bubble or curved film or membrane of the substance partially or completely enveloping or enclosing the expansion fluid can be generated. In some arrangements, the curved film or membrane of the substance can completely envelop the expansion fluid and can have a spherical shape. The surface(s) at the distal tip **130** of the cannula **114** (which can be, but is not required to be angled or beveled) can allow for a substance (which can be in the form of a bubble) to be removed from the cannula **114** and deposited on a target tissue (e.g., a retinal tear) with precision and with reduced risk of further tissue trauma. Additionally, any of the devices disclosed herein, including device **100**, can be configured to maintain the bubble on the distal tip when the gas stops flowing through the expansion fluid passageway, and/or release the bubble when the bubble can be brushed against a retina or when the gas flows again through the expansion fluid passageway.

The distal tip can have at least one surface (referred to herein as the support surface or end surface) **131** configured to support a substance for the generation of a film of the substance and, subsequently, a bubble of the substance. The end surface **131** or surfaces, if multiple as in some arrangements, can optionally be beveled relative to a longitudinal axis of the cannula body and/or at least one cannula exit port. The at least one end surface **131** can be configured to support a film of the substance (which can be a bioadhesive in any embodiments disclosed herein). The film of the substance can be manipulated by a fluid advancing through the fluid passageway **122** to change from a first shape or configuration to a second shape or configuration having a spherical or curved shape, or being in the form of one or more bubbles of the substance. For example, in some arrangements, second state of the film can be formed when expansion fluid is selectively advanced through the fluid passageway **122** to the at least one cannula exit port **132** and/or through a substance contained within the handpiece or cannula, thereby generating an approximately spherical or curved film of the substance. In arrangements where the substance is in the form of a bubble in the second state, the bubble can enclose completely around a volume of the expansion fluid.

The end surface **131** can be generally planar, as shown in FIG. 2. For example, FIG. 2 shows a side view and FIG. 3 shows a top view of a distal tip **130** comprising a flat beveled bubble port. The distal tip shown can also have a substance retention rim **133**. The exit port **132** can be on the same plane as the end surface **131**, and can be at an angle that is approximately 30 degrees, or approximately 40 degrees, or from approximately 30 degrees or less to approximately 45 degrees or more relative to a longitudinal axis extending through the cannula **114**. As shown in FIGS. 2 and 3, the entire rim of the cannula exit port **132** can be in approximately the same plane as the end surface **131**. The substance retention rim **133** of the end surface **131** can be configured to provide support for a bubble as the bubble forms, enlarges, and sits on the substance retention rim **133**. The substance retention rim **133** can be configured to keep the

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substance from coming off the distal tip **130** when the device is being navigated toward a target tissue surface or an eye.

FIG. 4A shows an example of a formation of a bubble **146** comprising a substance **150** filled with an expansion fluid **105** on the distal tip **130** of the device. A substance **150** is shown on a rim of the distal tip **130**. The bubble was formed by passing an expansion fluid **105** through the distal opening **132** of the device, which was covered by a film or layer of the substance **150**. As expansion fluid **105** is continued to be advanced through the distal opening **132**, the film or layer can stretch and move into a spherical or bubble-like shape, with more substance **150** being drawn into the bubble or the existing substance stretching as the bubble is enlarged, such as is shown in FIG. 4A, the substance **150** being used to form the bubble **146**. The bubble **146** is formed as expansion fluid **105** is advanced through the passageway **122** of the cannula **114** and through the distal port **132**. The expansion fluid **105** fills the interior space within the bubble **146**.

With the bubble **146** formed or enlarged to the desired size or wall thickness, the surgeon or user can then advance the distal tip **130** of the device having the bubble thereon to the desired treatment location, as shown in FIG. 4B. The distal tip **130** of the device can be used to advance the bubble **146** comprising the substance in a controlled fashion toward a defect **151** in the tissue or object surface **153**, as shown in FIG. 4B. Thereafter, the bubble **146** can be pressed in further contact with the defect **151** (which can be a retinal defect) or surface **153** of the object by advancing the distal tip **130** further toward the defect to elongate the bubble and spread the substance over the defect **151** and target surface **153**.

The bubble **146** and, hence, the substance **150** can thereafter be released from the distal tip in multiple different ways. For example, continued advancement of the distal tip **130** toward or against the target tissue surface and/or defect, or continued enlargement of the bubble, can cause the bubble to rupture against the target tissue surface. The distal tip can be made from a hydrophobic or other material configured to repel or reduce the surface tension of the substance relative to the distal tip so that at least a portion, or most or substantially all of the substance that was used to form the bubble can be deposited over the defect or against the target tissue surface. Additionally, as will be described below with reference to FIG. 43, an applicator device **1002** can be used to remove the bubble from the distal tip and/or apply or rupture the bubble against the defect or target surface. Further, in some arrangements, a movement of the distal tip relative to the target surface can exert a shear force on the bubble that can cause the bubble to burst or dissipate and deposit against the target surface. In some arrangements, a needle or other piercing element (both of which can optionally have a blunt or sharp end surface) or other suitable piercing instrument can be advanced either through the distal tip or separate from the distal tip toward the bubble to cause the bubble to burst against the target tissue surface. Further, any of the other devices or components described herein configured to facilitate the release of the bubble from the distal tip can be used to facilitate the release of the bubble from the distal tip.

Optionally, the distal tip **130** of the device can be moved in close proximity to the defect or target location before the bubble is fully or even partially or substantially expanded so that the bubble can merely be expanded against the defect or target surface. Thereafter, as shown in FIG. 4D, the distal tip **130** can be retracted or withdrawn from the target location, leaving the substance **150** deposited against the target surface **153** so as to cover the defect **151**.

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Any arrangements of the device or methods disclosed herein can include a liquid reservoir or chamber, together with means for applying the liquid from said reservoir to the film retaining member (a ring or loop such as a flexible loop) or the distal end of the cannula to form a film thereon, whereby each time the device can be actuated and/or touched to the retina to remove the film, additional liquid to provide a new film may be applied to said ring to make the device ready for next use, by retracting the ring.

In any arrangements, the substance can be provided from reservoir that can be in the form of a cartridge, absorbable material, or chamber which may be detachably mounted on the device, integral to the device, or otherwise, or can be advanced into an internal passageway of the device, such as the expansion fluid passageway. The device can optionally be configured such that, when said cartridge or chamber is emptied, it may be readily replaced with a full cartridge, absorbable material or chamber of duplicate construction.

Any arrangements disclosed herein can be configured to have a substance supply lumen or chamber within the handle portion and/or the cannula, or advanceable within the handle and/or cannula such as but not limited to cannula **114**. The substance supply channel can be configured to advance the substance to the distal tip of the cannula, such as cannula **114**. In some arrangements, as shown in FIGS. 5 and 6, the substance supply channel can have a separate elongate body portion that can be advanced distally through the expansion fluid passageway of the cannula to the distal tip of the cannula so that substance can be ejected out of the end of the substance supply channel onto the opening in the distal tip of the cannula to provide and/or replenish the substance that is on the distal tip for formation of bubble elements. The substance supply channel can optionally be advanceable and retractable within the expansion fluid passageway of the cannula as desired. In some arrangements, as shown in FIG. 7, one or a plurality of substance supply channels can be formed in a wall portion of the cannula so that a steady supply of the substance can be provided to the distal tip without requiring a separate device for doing so.

With reference to FIGS. 5 and 6, in some arrangements, the substance supply channel or tube **158** can have a separate elongate body **160** that can be advanceable within the expansion fluid passageway of the cannula (not shown in FIG. 5). The elongate body **160** can have a passageway **162** extending through the length of the elongate body **160** which can be in fluid communication with a distal opening or port **163**. The distal port **163** can extend through the distal end **164** of the supply channel **158**. The body portion **160** can therefore be sized and configured to be advanceable without restriction through the expansion fluid passageway of the cannula with which the supply channel **158** is configured to work with. For example, the elongate body **160** of the supply channel **158** can be approximately 20% smaller than the inner diameter of the expansion fluid passageway of the cannula, or from approximately 2% to approximately 20% or more, or from approximately 5% to approximately 10% smaller than the inner diameter of the expansion fluid passageway of the cannula. The distal end **164** of the supply channel **158** can have a shape and angle that approximately matches the shape and angle of an end portion of the inside surface of the inside of the expansion fluid passageway and internal surfaces of the inside of the cannula, and a size that is slightly smaller than the expansion fluid passageway.

The supply channel **158** can be advanced distally within the expansion fluid passage inside the cannula of the bubble generating devices so that the distal end **164** is adjacent to or in contact with the end of the internal passageway.

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Thereafter, the substance can be advanced through the one or more openings **163** to create a film across the distal port in the bubble generating device. The substance can be spread across the distal port in this fashion. The supply channel **158** can be used in this fashion to fill the tip of any cannula or device disclosed herein with the substance. The end surface may have multiple substance delivery holes (as shown in FIG. 7), or a single hole **163** as shown in FIGS. 5 and 6. Each hole may be large, for example and without limitation encompassing greater than 50% of the end surface **163** of the tip of the substance supply channel, or small, for example and without limitation encompassing 5-25%, or less than 50% of the end surface **164** of the beveled tip of the substance supply channel. Some arrangements can have five or more holes, or from 4 to approximately 10 or more holes in the distal end of the supply channel.

Some arrangements of the system, device, cannula, or distal tip comprise a space beneath the substance retention rim **133** and the substance supply channel **158**. Said space can be configured to allow the bioadhesive substance to coalesce from the substance supply channel **158**.

As mentioned above and as shown in FIG. 7, any arrangements of the supply channel disclosed herein can include a substance supply tip comprising any desired number of openings **163** at a distal end portion thereof, including one opening, two openings, three openings, four openings, five openings, six openings, seven openings, or up to twenty or more openings, or between five and ten openings, or any number of openings between any of the foregoing values. As mentioned, the substance supply channel can include a proximal end that extends into the fluid passageway of the cannula. The substance supply channel can be sized so that the expansion fluid can flow around an outside surface of the substance supply channel when the expansion fluid is blown through the fluid passageway of the cannula.

In some arrangements, the cannula can optionally include an outer wall and an inner wall, the inner wall comprising the substance supply channel, and a space between the inner wall and the outer wall comprising the expansion fluid passageway of the handpiece, the fluid passageway of the cannula body, or a fluid passageway of the cannula base or tip. The substance can optionally be advanced through the innermost lumen or inside the inner wall. The substance supply tip can include a surface that can be beveled as compared to a longitudinal axis of the substance supply channel body and the substance supply exit port, per any of the details of the other beveled tips disclosed herein or otherwise.

Another example of a cannula **170** having an internal substance supply passageways **172** is shown in FIG. 8. The cannula **170** can have a plurality of substance supply passageways **172** that can, optionally, be integrally formed with the cannula, extending through the cannula **170** that are configured to permit passage or advancement of the substance from a proximally located source of substance or substance reservoir to the opening **176** in the distal tip **178**. The arrows through the substance supply passageways **172** in FIG. 8 illustrate a direction of flow of the substance through the substance supply passageways **172** when the surgeon or user advances the substance through the substance supply passageways **172**. The substance supply passageways **172** can optionally be in communication with a manifold **180** surrounding the opening **176** in the distal tip **178**. The manifold **180** can be sized and configured to create a film or membrane of the substance across the opening **180** in the distal tip **178** when the substance is advanced through the passageways **172** so that, when an expansion fluid is

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advanced through the expansion fluid passageway **184** of the cannula **170**, a bubble **186** can be formed from the film or membrane of the substance. The manifold **180** can have a series of radially inwardly projecting openings, an open annulus, openings or jets that can be configured to facilitate the formation of the film of substance. The cannula **170** can be formed as a unitary piece, or from multiple pieces coupled together. The opening **176** can be sized and configured to ensure the consistent formation of bubbles when an expansion fluid is passed through the substance, and/or the consistent formation of a membrane of the substance supplied through the passageways **172**.

As described, any arrangements of the devices disclosed herein can be configured so that the substance supply channel surrounds at least a portion of the expansion fluid passageway of the handpiece, the fluid passageway of the cannula body, and/or a fluid passageway of the cannula base or tip. As illustrated in FIG. 9, for example, device **200** can have an elongate body portion **202** having an expansion fluid passageway **204** extending therethrough. The expansion fluid passageway **204** can be partially or completely surrounded by a substance supply channel or lumen **206** that can be used to advance the substance to the distal tip **210** of the device **200**. The lumen **206** can be a single annular lumen or can comprise one or a plurality of openings all in communication at a proximal and a distal end or portion of the lumen **206**. In some arrangements, a deflector or deflectors **212** can be positioned at a distal end of the lumen **206** and be configured to direct the flow of substance radially inwardly toward the opening **216** in the distal end of the device **200**. The deflector can optionally have an annular shape. The deflector can be angled inwardly at an angle that is less than approximately 90 degrees, or have a ramped surface to provide a more efficient flow pathway of the substance toward the opening **216**.

The device **200** can optionally include a continuous (360 degree) circumferential deflector member **212** or a plurality of discrete deflector members. The deflector member can be configured to allow for longer retention of the bubble on the tip of the cannula. This configuration allows fluid to flow around the expansion fluid (which, in any arrangements disclosed herein, can be air, sterilized air, or another inert gas, or the other fluids described herein) and form a film of substance at the cannula exit port **216**. After the film has been formed, the expansion fluid can be advanced against the film to cause a bubble to be formed. FIG. 10 illustrates a bubble being formed from the film of substance advanced through the substance supply channel or lumen **206**. The bubble may be supported by a flared opening.

In other arrangements, the substance supply channel can include a tube or cannula that can be advanced within the expansion fluid passageway **362** of the handpiece, the fluid passageway of the cannula body, or a fluid passageway of the cannula base or tip. The substance supply channel tube can be advanced to be in contact with or within the distal tip toward the cannula exit port. The substance supply channel tube can have a distal end that reaches near the cannula exit port.

Any arrangements of the system and/or device disclosed herein can further have a second substance supply channel. The substance supply channel and the second substance supply channel each can include a separate bioadhesive substance, for example a polyethylene glycol solution and a trislysine amine solution, hydrogel, double layer hydrogel, or polymeric hydrogel.

The at least one end surface can be configured to support a bioadhesive bubble. The end surface, retention ridge, lip,

or rim can include a flat surface configured to hold and/or maintain a bioadhesive bubble. The distal tip and/or the end surface of the distal tip can include a retention ridge, lip, or rim comprising a flat surface configured to hold and maintain the bioadhesive bubble.

The end surface of the distal tip can have a 1-15°, 15-30°, 30-45°, 45-60°, 60-75° angle relative to a square end (in either direction), or an angle in a range comprising more than one of the aforementioned angle ranges, compared to the longitudinal axis of the cannula body and/or at least one cannula exit port. The angle of the end surface can be configured to enhance retention of the bioadhesive bubble so the bubble can be kept from falling off.

The distal tip of some arrangements can be configured to not have a bevel. For example, the distal tip may comprise a flat or square surface with a cannula exit port at **900** in relation to the longitudinal axis of the cannula body, or the distal tip may be rounded and/or blunt without a flat surface.

Any arrangements of the distal tip and/or the end surface of the distal tip can include a retention ridge, lip, or rim for retaining or supporting a bubble thereon. Additionally, the size, angle and/or shape of the distal tip (such as distal tip **130** or any other distal tips disclosed herein) and the retention ridge, lip, or rim thereof can be based on the application and on other factors related to the substance, including without limitation the viscosity of the substance. For example, the substance retention rim **133** may be thinner (e.g. 1 mm) for a viscous substance than for a non-viscous substance (e.g. 2.5 mm). In other arrangements, the substance retention rim **133** can be thicker when configured for a substance with lower viscosity than for a higher viscosity substance. The cross-sectional shape of the port **132** can optionally be round, square, ovalar, rectangular, triangular, pentagonal, hexagonal, star shaped, or otherwise. The shape can be selected based on the viscosity of the bioadhesive substance and the size or other parameters of the desired bubble or bubbles, or foam to be formed.

The end surface **131** can be flat, curved, or otherwise, and can have a depression formed therein sloping toward the opening **132**. The depression can be angled inward and can have a flat cross-sectional profile or be curved. Any arrangements disclosed herein can have a plurality of openings **132** formed in the distal end portion to enable the simultaneous formation of a plurality of bubbles of the substance, or to form a foam-like structure of the substance. Any arrangements can have two or more openings, or from two to approximately twenty or more openings, or from four to twelve openings, or from and to any values within these ranges.

FIGS. **11** and **12** illustrate another arrangement of a cannula **250**. The end surface **252** of the distal tip **254** of the cannula **250** can optionally be circular, ovalar, non-circular or otherwise. For example, the distal tip **254** can optionally be round, square or rectangular shaped. Additionally, the distal tip **254** can have a non-planar end surface **252** such as is shown in FIGS. **11** and **12**. The end surface **252** can be angled inward toward a center of the cannula **258**. The end surface **252** can have a bevel or angle formed around the opening **256**. The opening **256** can be in fluid communication with an expansion fluid passageway **258** through the cannula **250**. The end surface **252** can be at an angle A (as shown in FIG. **12**) that is approximately 45 degrees relative to the longitudinal axis C (as shown in FIG. **12**), or from approximately 30 degrees or less to approximately 90 degrees or more, or from approximately 40 degrees to approximately 70 degrees, or from approximately 50

degrees to approximately 60 degrees relative to the longitudinal axis C, or to and from any values within these ranges.

Additionally, the cannula **250** can have a hub or hub portion **260** and connector threads **262** on an external surface (as shown) or internal surface thereof. The cannula can optionally include a solid cannula wall, a core or cannula fluid passageway **258** through which a substance supply channel or an expansion fluid can be advanced, an opening or cannula exit port **256** at the tip of the cannula, and a space **264** configured to maintain a film of substance such as a bioadhesive substance. As with any arrangements disclosed herein, the cannula **250** can be dipped into a substance dispenser as described below for adding the substance to the distal tip of the cannula, and can retain the bioadhesive substance as a film in the space **264** at the tip.

Accordingly, in any arrangements, the end portion of the cannula (whether beveled, square, or otherwise) can have a concave or inwardly curved depression formed in the end surface thereof. Alternatively, in any arrangements, the end portion of the cannula (whether beveled, square, or otherwise) can have a convex or outwardly curved depression formed in the end surface thereof. The end surface may be flared or flanged at the end, and/or include a space that holds or provides greater surface area to the bioadhesive substance. An advantage of the flared or flanged surface is that it can retain a film of the bioadhesive substance that forms a bubble when a gas such as air blows into and/or against the film.

Some arrangements of the distal tip can have multiple end surfaces. Some arrangements of the distal tip can have no more than one end surface, while in other arrangements, a plurality of end surfaces are used. The distal tip can be blunt or rounded, or can include a blunt or a rounded end.

The tip can include one or more grooves at, for example, the distal tip. An advantage of having grooves at the distal tip is that the one or more grooves may help retain bioadhesive substance and aid in bioadhesive substance bubble formation. The distal tip can include one to ten, or from three to six (for example, five) grooves, or any number of grooves within that range or more in any of the same numbers. The distal tip can include an inner surface comprising the grooves. For example, FIG. **13** illustrates a side view of another arrangement of a cannula **300** having a plurality of grooves **302** formed on an inside surface of the distal tip **304**. The distal tip **304** can optionally include an outer surface **306** also or alternatively having one or more grooves **310**, as illustrated in FIG. **14**, which is a side view of a distal tip having a plurality of grooves **310** on an outside surface of the distal tip. Any arrangements can have ridges, grooves, or other patterns of protrusions and/or depressions on the inner or outer surface of the tip of the cannula, optionally including the end surface, to help retain substance within and/or on the tip of the cannula. The ridges, grooves, or other patterns may be configured to retain the bubble or hold the bubble in place.

In any arrangements, the cannula, cannula base, cannula body, and/or distal tip can have a cross sectional diameter that is from approximately 0.1 mm to approximately 10 mm or more, or from approximately 1 mm to approximately 4 mm, or from approximately 1.5 mm to approximately 2 mm, or to and from any values within these ranges. In some arrangements, the cannula, cannula base, cannula body, and/or distal tip can have a cross sectional diameter that is from 0.1-0.5 mm, 0.5-1 mm, 1-2 mm, 2-3 mm, 3-4 mm, 4-5 mm, 5-6 mm, 6-7 mm, 7-8 mm, 8-9 mm or 9-10 mm. The cannula, cannula base, cannula body, distal tip and/or cannula exit port can have a length of approximately 0.1-0.5

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mm, 0.5-1 mm, 1-2 mm, 2-3 mm, 3-4 mm, 4-5 mm, 5-6 mm, 6-7 mm, 7-8 mm, 8-9 mm, 9-10 mm, 10-20 mm, 20-30 mm, 30-40 mm or 40-50 mm. The cannula, cannula base, cannula body, distal tip and/or cannula exit port can include an external diameter, an internal diameter and a width between the internal diameter and the external diameter, each being one of 0.01-0.05 mm, 0.05-0.1 mm, 0.1-0.5 mm, 0.5-1 mm, 1-2 mm, 2-3 mm, 3-4 mm, 4-5 mm, 5-6 mm, 6-7 mm, 7-8 mm, 8-9 mm, 9-10 mm, 10-20 mm, 20-30 mm, 30-40 mm or 40-50 mm any of the components or features disclosed herein can have any suitable or typical size, including without limitation any size within any ranges stated herein.

The retention edge, ridge, or lip can be 0.1-0.5, 0.5-1, 1-2, 2-3, 3-4, 4-5, or 5-10 mm thick. The retention edge, ridge, or lip, the retention edge, ridge, or lip can be 0.1-0.5, 0.5-1, 1-2, 2-3, 3-4, 4-5, or 5-10 mm long. Some arrangements include a retention edge, ridge, or lip that can be set back a short distance from the very tip of the cannula. The distal tip may or may not have a central air tube.

In any arrangements, the opening can be at the very tip of the distal end. Optionally, the opening can be on the side of the tip of the distal end. The opening can be on the side of the distal end, but more proximal than the tip of the distal end.

Some arrangements of the system and/or device include circumferential or non-circumferential intermittent support structures. The intermittent support structures can connect the outer wall to the inner wall. Some arrangements can include 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, or from 4 to 10, or any number of circumferential and/or non-circumferential intermittent support structures. Some arrangements comprise circumferential or non-circumferential intermittent support structures connecting the outer wall and the inner wall, for example 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, any number therebetween, or more circumferential and/or non-circumferential intermittent support structures. FIG. 15 shows an example of a distal tip 350 comprising intermittent attachments or support structures 352 between a central tube 354 and an outer tube 356. The support structures can be configured to permit fluid passage around such structures.

The handpiece or cannula can include an outer wall and an inner wall, and a space such as space 358 of cannula 350 between the inner wall and the outer wall. The space between the inner wall and the outer wall can comprise a substance supply channel. The inner wall can bound the expansion fluid passageway 362 of the handpiece, the fluid passageway 362 of the cannula body, or a fluid passageway 362 of the cannula base or tip.

In any arrangements, the expansion fluid can be provided from a source separate from the device, or from a reservoir or cartridge or absorbent material within the device, or built into the device. As described, any arrangements can have a fluid passageway through which gas can be advanced from the source. The device can optionally have a connector such as a port at the proximal end to which tubing or other conduit may be attached to allow pressurization of contents (such as the expansion fluid) within the device, in the presence or absence of a plunger or mobile plug. The handpiece or device in any arrangements disclosed herein can include an expansion fluid actuator. The expansion fluid actuator can be coupled with a proximal end of the device so as to be in fluid communication with the expansion fluid passageway. The expansion fluid actuator can be configured to be activated or moved by a user. The moving component can be configured such that an actuation of the moving component by the user causes flow or movement of the expansion fluid through the

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expansion fluid passageway, such as from the proximal end to the exit port of the distal end. The handpiece can include an expansion fluid actuator connected at the proximal end to the expansion fluid passageway, and configured to be engaged by a user, wherein said engagement can cause movement of the expansion fluid through the expansion fluid passageway from the proximal end to the exit port of the distal end.

The expansion fluid actuator can optionally include a roller wheel or roller ball that can be moveable forward and backward in order to advance or withdraw, respectively, a fluid through expansion fluid passageway. In this configuration, a roller wheel device can include a plurality of movable ribs or spokes coupled with a cylindrical outer surface that are configured to rotate about a center axis. In any embodiments disclosed herein, the roller wheel can be configured to expel the expansion fluid from the bladder through the expansion fluid passageway from the proximal end to the exit port of the distal end when the roller wheel is rolled or moved along a length of the tube or chamber by, for example, a person's finger, a small motor, a coil spring, or otherwise.

A non-limiting example of an expansion fluid actuator 500 that can be used with any of the device arrangements disclosed herein is shown in FIGS. 16A and 16B. The actuator 500 can have a roller wheel 502 configured to move along a length of the expansion fluid conduit 504 and to squeeze the expansion fluid conduit 504 so as to flatten and/or collapse the expansion fluid conduit 504 as the roller wheel 502 moves along the length of the expansion fluid conduit 504. As this occurs, the roller wheel 502 can force fluid within the passageway 506 in the distal direction, toward the distal tip of the cannula where the fluid can contact a film of substance such as a bioadhesive and create a bubble. The conduit 504 can be made from the resilient tubing material or even a non-resilient single use tubing material that collapses as the roller wheel 502 advances along the length of the conduit 504. In any arrangements, the roller wheel 502 can roll in both a distal direction and a proximal direction, withdrawing air back into the passageway 506 as the roller wheel is moved in the proximal direction. The actuator 500 can optionally also include a second roller coupled with the first roller wheel that can be joined with an roll together with the first roller so as to exert an opposing force on the tubing that squeezes the tubing between the first and second rollers. Some arrangements include a roller ball that can be moveable forward and backward, or in any restricted or unrestricted fashion 360 degrees around a point or axis.

Some arrangements include a fixed device where a bladder moves relative to a ball, sphere, stopper, plunger, or other stoppage device. For example, some such arrangements include a port in the handle configured to receive an adhesive substance into the handle, where a bladder containing the adhesive can be configured to be moved from a distal to a proximal direction, and the ball creates a barrier to the contents of the bladder, thus causing expulsion of the bioadhesive through the distal end of the device. Some arrangements include a bioadhesive packet that may be loaded into the device.

As described, the handpiece can further have a source of an expansion fluid, which can include an expansion fluid actuator, connected at the proximal end to the expansion fluid passageway. The expansion fluid actuator can be configured to be engaged, directly or indirectly, by a user, wherein such engagement can cause movement of the expansion fluid through the expansion fluid passageway

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from the proximal end to the exit port of the distal end. In some arrangements, the expansion fluid actuator can include a compressible bladder, syringe, a collapsible vial or other reservoir, such as an accordion type collapsible reservoir, or other similar device configured to expel the expansion fluid from the bladder through the expansion fluid passageway from the proximal end to the exit port of the distal end of the device. In some arrangements, the expansion fluid actuator can have a roller wheel moveable forward and backward, and be configured to expel the expansion fluid from the bladder through the expansion fluid passageway from the proximal end to the exit port of the distal end.

In still additional arrangements, the expansion fluid actuator can have a sliding piece, such as a plunger, etc. In some arrangements, the expansion fluid actuator can include a syringe filled with an expansion fluid, such as air. Optionally, the expansion fluid actuator can comprise a compressed fluid or gas cartridge, such as a compressed air cartridge that can be fluidically coupled with the expansion fluid passageway. A trigger, valve, stopcock, or other device or component can be used to control the flow of air from the compressed air cartridge into the expansion fluid passageway.

The expansion fluid actuator can optionally include a compressible bladder. The compressible bladder can be configured to expel the expansion fluid from the bladder through the expansion fluid passageway from the proximal end to the exit port of the distal end. The expansion fluid actuator can include a compressible bladder configured to expel the expansion fluid from the bladder through the expansion fluid passageway from the proximal end to the exit port of the distal end. The compressible bladder can have a diameter that extends out from the center of the handle 1-1.5, 1.5-2, 2-3, 3-4 or 4-5 times a diameter of another portion of the handle such as the elongate body, on one or more sides of the handle or elongate body. For example, the compressible bladder may extend 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 10-15, 15-20 or more from a side of the handle or elongate body of the handle. The compressible bladder can include any flexible material such as rubber, plastic, a flexible metal, or an elastic material.

FIG. 17 shows a side view of an embodiment of an expansion fluid actuator 520 that can be used with any of the device arrangements disclosed herein to provide an expansion fluid to a distal end or tip of the cannula and, hence, to the substance to form a bubble. The arrangement of the fluid actuator 520 illustrated in FIG. 17 can include a compressible bladder 522 having an internal air space 523 that is in fluid communication with an expansion fluid passageway 524 of and expansion fluid channel 526. The air bladder 522 can extend or protrude away from one side of the handle such as from the top of the handle so that a user can exert a force F in generally one direction so as to collapse the air bladder 522 and to force an expansion fluid to advance through the expansion fluid passageway 524. The actuator 520 can have a support or support surface for the compressible bladder on the bottom of the compressible bladder. The support for the compressible bladder may comprise the handle of the device or system, or an intermediate support structure, depending on the arrangement. The air bladder can be compressible, for example, with a person's finger to force air from the bladder.

Some arrangements can include an actuator mechanism provided with a compression device comprising a normally expanded, resilient air storage bulb, thus forcing air distally through an air tube toward a distal tip of a cannula or through a bubble-making ring at the end of the device. Some

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arrangements can include a trigger mechanism that compresses the air storage bulb from one side. Some arrangements include a trigger mechanism that compresses the air storage bulb from more than one side simultaneously. The bulb may be made of a material which can be sufficiently resilient to return the trigger to its normal position, but if desired a small coiled spring may be associated with the trigger to maintain the trigger in a normal inactive position and to return the same to such position after each operation.

The device can include a cannula and an associated distal tip, separate from the bubble generating portion of the device, that permits the blowing of air onto the retina to create a drying effect, prior to application of the bubble. In some arrangements, the device can be configured to permit the device to blow air through the distal tip of the cannula without the presence of a substance so that air can be blown onto the target tissue.

FIG. 18 shows another arrangement of an actuator 540 that can be used to advance the expansion fluid toward the distal end of a cannula of any of the arrangements of devices disclosed herein. In any arrangements, the actuator 540 can have a bladder 542 having the volume of space 543 therein that is in fluid communication with the passageway 544 through an expansion fluid conduit or channel 546. The bladder 542 can be arranged with flexible ribs or struts, corrugations, or otherwise to improve the flexibility and collapsibility of the bladder. The ribs can be configured to help the bladder 540 retain its shape and elasticity, and/or to bias the bladder to return to an expanded shape. The ribs can be made of the same material as another portion of the compressible bladder, or as the rest of the compressible bladder, or may be made of another flexible material.

In any arrangements, the actuator 540 can have one or more one-way valves in fluid communication with the bladder so that air can refill the bladder when the bladder is released and allowed to expand back to its original expanded state, and to prevent air from sucking back from a distal portion of the conduit 546 back into the bladder 542. The bladder 542 can be sized to have any desired volume of expansion fluid contained within the bladder 542. The bladder 542 may be squeezed or otherwise compressed in any direction to expel the expansion fluid toward the distal tip.

In some arrangements, the bladder 542 can be spherical or generally spherical and extend in all directions radially away from the centerline of the conduit 546. In other arrangements, the bladder 542 can have a semi-spherical shape, or any other desired shape.

Any arrangements of the systems and/or devices disclosed herein can include a separately or electrically powered means for advancing the expansion fluid through the expansion fluid passageway of the handpiece or through the expansion fluid passageway of the cannula. For example, the expansion fluid passageway can include a proximal end connected to a tube, channel, or other passageway connected to and configured to receive expansion fluid from an expansion fluid receptacle. The receptacle can include a valve. Said valve can be configured to open by operation of a button on the handpiece of the system or device, or by operation of a foot pedal. Said valve can be configured to open by manual operation such as through operation of a screw handle.

In any arrangements, the expansion fluid actuator can include a sliding piece. The sliding piece or other expansion fluid actuator can be configured to move a plunger, such as a plunger toward the distal end of the device. Movement of the plunger can force the expansion fluid through the expansion

sion fluid passageway and/or exit port of the handpiece, and/or through the cannula expansion fluid passageway and/or cannula exit port of the cannula to produce the bubble.

FIGS. 19 and 20 show a non-limiting example of an expansion fluid actuator 560 comprising a sliding piece 562. The sliding piece 562 can include a moveable part or slider 562, where the moveable slider 562 can be coupled with a sealed diaphragm, a piston, a plunger or other similar component located within the chamber 570 of the actuator 560 that can move with the movement of the slider 562 so as to change a volume of space within the chamber 570. The chamber 570 can optionally have a general cylindrical shape at least in a middle portion thereof. The slider 562 can be configured to be movable along a pathway or groove 564 from a proximal end 566a to a distal end 566b of the device, thereby expelling air or expansion fluid through the expansion fluid passageway of the device or system. It shall be appreciated that the slider 562 can be moved in both directions, thus enabling control of the amount of expansion fluid expelled from the device.

In some arrangements, moving the slider 560 toward the proximal end of the device can be configured to withdraw air from the passageway that the chamber 570 is connected to, so as to reduce the volume of a bubble or otherwise withdraw a fluid from the fluid passageway of the cannula. In some arrangements, one or more one-way flow valves can be positioned in fluid communication with the chamber to prevent withdrawal of air from the cannula as the slider 560 is moved toward the proximal end of the device however. The chamber can be rechargeable using sterile or filtered ambient or another fluid source as the slider is moved proximally or from a separate source.

The device 560 can be handheld. The distal end 566b can be tapered. When the slider 560 is moved from the proximal end 566a toward the distal end 566b, fluid within the chamber 570 can be forced through the device to put pressure on a film of substance, that may be liquid, gel or some other non-solid composition, that can be suspended across an opening in the distal end 566b. Thereby, a bubble of the substance can be formed out of the hole in the tapered distal end 566b. The film of substance can be formed across a ring that can be positioned inside the distal end of the device. In other arrangements, fluid (e.g., a gas) can be forced from the chamber 570 and into a cannula of any other arrangements disclosed herein toward a distal tip of the cannula by moving the slider 562 toward the distal end 566b of the device.

Some arrangements can optionally include a mechanism for locking a trigger mechanism once the bubble is created, allowing maneuvering of the device in the eye without the need to maintain pressure on the trigger to avoid deflation of the bubble. Additionally, in some arrangements the diaphragm, plunger, or piston within the chamber 570 or any other chamber, such as a cylindrical chamber formed as part of the handle portion or attachable thereto, can be moved by a rotating or twisting action such as by spinning a dial or other threaded components to move the diaphragm, plunger, or piston within the chamber and, thereby, force the expansion fluid out of the chamber and into the cannula. The device of this arrangement, because of the threaded nature of the actuator, can resist inadvertent movement by the back pressure provided by the bubble and also provide greater resolution and control by the user in terms of the amount of air that is evacuated from the chamber.

In some arrangements, both the proximal or the distal end can optionally include an opening that may produce bubbles.

Optionally, either the proximal or the distal end, but not both, can include an opening for the production of bubbles. Both the proximal and/or the distal end can include multiple openings that may produce multiple bubbles at the same time.

Any of the arrangements of the devices disclosed herein can include a one-way flow valve that can be configured to allow fluid, such as the expansion fluid or the substance, to flow in the distal direction when the pressure on the proximal side of the valve is greater than on the distal side of the valve, and to prevent back-flow (i.e., flow in the proximal direction) of the fluid or substance.

Some arrangements include a mechanism for locking the actuator mechanism once the bubble is created, or to prevent any backflow of fluid toward the proximal end of the device once the bubble is created. In this state, the device can be maneuvered and manipulated by the surgeon (such as, optionally, in the eye) without the need to maintain pressure on the trigger to avoid deflation of the bubble.

The distal tip of any arrangements disclosed herein can include a one or more flaps adjacent to or covering the distal port of the device. For example, the distal tip can include four flaps, or from two to eight or more flaps, or from three to six flaps. The flaps can be connected to a circumferential edge at or adjacent to the distal tip of the device and/or extend across the cannula exit port.

FIGS. 21 and 22 provide illustrate another arrangement of a device 590 having an opening 592 extending through a distal surface 593 of a distal end of the device, the opening 592 being in fluid communication with a passageway 594 extending through the device. One or more bendable, resilient flaps 596 (four being shown) selectively cover the opening 592. The flaps 596 can be made from silicon and can form a diaphragm over the opening. The flaps 596 can serve or act as a valve. As shown in FIG. 22, the flaps 596 may include slits 595, rounded or beveled edges, and/or a hole. One or more of the flaps can optionally each comprise a beveled edge.

The silicone diaphragm with one or more slits and/or a central hole may be used with any of the various cannula or applicator tips described herein. The silicone diaphragm may function as a valve to hold substance inside, and/or to control bubble formation. The optional slits or hole at the center of the silicone diaphragm may be any shape. The flaps comprise rubber, silicone, plastic, or an elastic material.

The handle of any arrangements disclosed herein can have a joining portion. The joining portion can include a coupling end. The coupling end can be configured to reversibly or non-reversibly couple with a proximal portion of any cannula arrangement or elongate body arrangement disclosed herein. The proximal portion of the cannula can be configured to reversibly couple with the coupling end of the elongate body or the joining portion of the handle. If the handle has an elongate body, the elongate body can include a joining portion, the joining portion comprising a coupling end configured to interact with a proximal portion of the cannula.

The handle, cannula, or any component thereof or other components of the device, can be made from or can comprise plastic, metal, polyvinyl chloride, glass, acrylic, carbon fibers and/or wood. The distal tip plug or protective tip cover can include plastic, metal, polyvinyl chloride, glass, acrylic, carbon fibers and/or combination thereof or any other suitable materials. The handpiece or cannula, or a component thereof, can be minimally, moderately, or maximally thermally conductive.

The handpiece elongate body, cannula, exit port and/or expansion fluid passageway can have an approximately 20 mm to approximately 50 mm cross-sectional diameter, size, or width, or an approximately 25 mm to approximately 40 mm cross-sectional diameter, size, or width, or a 0.1-0.5 mm, 0.5-1 mm, 1-2 mm, 2-3 mm, 3-4 mm, 4-5 mm, 5-6 mm, 6-7 mm, 7-8 mm, 8-9 mm, 9-10 mm, 10-20 mm, 20-30 mm, 30-40 mm or 40-50 mm cross-sectional diameter, size, or width. The handpiece or handle can have a length of from approximately 4 cm to approximately 12 cm or more, or from approximately 6 cm to approximately 9 cm, or from and to any values within these ranges. The cannula can have a length of from approximately 0.5 cm or less to approximately 10 cm or more, or from approximately 2.5 cm to approximately 5 cm.

Additionally, in any arrangements, infusion and/or aspiration of the bioadhesive substance and/or an activation solution can be manual or foot pedal controlled. Therefore, any arrangements disclosed herein can include a foot pedal. The foot pedal can be configured to expel air or expansion fluid through a tube or channel into the expansion fluid passageway so that it may flow toward and through the distal portion of the device to form a bubble at the distal tip of the device.

A liquid retaining member in the form of an annular wire ring can be used with any of the cannula arrangements disclosed herein. The ring can optionally be secured by means of an angular arm to the tip of the distal end of the device. The liquid retaining member (also referred to herein as a ring) need not be an annular ring but could take any desired shape. The liquid retaining member can be adapted to retain a bubble forming liquid thereacross. When the liquid is applied to the ring, a film of said liquid can be formed thereacross and can be retained in such position until displaced therefrom by air or other pressure.

In some arrangements, an angular arm can be made integral with the ring. The arm can extend along part of the length of the body of the device. The angular arm of the ring can be attached to the device using any suitable components or means. The ring have an attachment, that may be shaped like a button or any other suitable shape, that may extend outside the central core of the device so that it can be easily manipulated with a finger or fingers. The attachment may be used to move the angular arm and attached ring distally and proximally.

Some arrangements include a ring or plurality of rings that provide the means to retain a film of material across the space within the ring, capable of forming a bubble or bubbles when a stream of air or gas is directed against said film to form bubbles. In some arrangements, the ring can be fixed and can be dipped in liquid to create a film across the ring. Some arrangements include a fixed ring that can have liquid actively distributed across the central aperture of the ring. The ring can be a hollow reservoir, that can hold fluid, and that has one or more openings on the inner aspect of the ring that can allow fluid to be infused through the hollow ring then exiting the ring through the openings in the inner aspect of the ring in such a way that creates a film of the liquid across the ring aperture, enabling a bubble to be formed as gas passes through the ring.

For directing a stream of air against the film which can be retained across the ring, an air tube can be mounted axially within the device. The spacing between the ring and the end of the air reduction tube controls the action of the bubble or bubbles which are formed. Also the volume, as well as the velocity of the air passing through the tube, which velocity can be controlled by the bore of the outlet end of said tube,

may have an effect on the results obtained. If the end of the tube is too close to the ring, the force of the air stream may merely displace the liquid film without forming the bubble or bubbles. This can cause the rupture of the film. On the other hand, if the spacing between the ring and the end of the tube is too great, the air stream may be dissipated before it reaches the film and bubbles will not be formed. By optimizing the spacing between the air tube and the ring, a bubble or series of bubbles, the size of which can be controlled by the size of the retaining ring, will be formed upon depression of the bulb or a single large bubble may be formed upon each operation of said bulb. Thus, by adjusting the distance between the ring and the tube, as well as by controlling the size of the bore of the tube, a desired effect may be obtained.

Some arrangements of the system and/or device can include a loop. The loop can be retractable—e.g., the loop extends from an end of the device, such as the distal end, and can be retracted in a proximal direction partially or completely within the distal tip of the device to change the size and/or shape of the loop. The loop may be retracted into a handle, cannula or distal tip in accordance with some arrangements. The loop can include a wire. The loop or wire can include a metal such as iron, steel or stainless steel, titanium, nitinol, platinum, or another material such as plastic. The ring can be configured such that the substance fills or forms a film on the wire loop. The adhesive may fill or form a film on the wire loop when the wire loop is retracted and/or expanded, depending on the arrangement. The film may come out from the distal tip and expand or eject into the loop.

The system or device can be configured so that after the bioadhesive substance forms a film on the flexible loop, the wire loop may be applied to a retinal break or tear. Some arrangements include a method of applying a flexible loop with a film of bioadhesive substance to a retinal break or tear. When the wire containing a film is laid on a surface such as a retina or retinal tear, the device and substance can be configured such that the film adheres to the surface (to a degree greater than the adherence of the film to the loop) and can be displaced from the loop.

The loop can be non-flexible and/or non-retractable. The wire or loop can include a thickness of approximately 0.02 mm to approximately 1 mm, or 0.01-0.02 mm, 0.02-0.05 mm, 0.05-0.1 mm, 0.1-0.5 mm, 0.5-1 mm, 1-2 mm, 2-3 mm, 3-4 mm, or 4-5 mm, or a range comprising any combination of said thicknesses. For example, the loop may optionally comprise a flexible suture material such as a polypropylene or collagen suture, and may have one of the aforementioned diameters. Suitable sutures include but are not limited to sutures having a USP designation of 11-0, 10-0, 9-0, 8-0, 7-0, 6-0, 5-0, 4-0, 3-0, 2-0, 0, 1, 2, 3, or 4. The loop can have a diameter of 1-2 mm, 2-4 mm, 4-6 mm, 6-10 mm, 10-20 mm, or 20-30 mm, or a range comprising any combination of said diameters.

FIG. 23 illustrates an example of a system and/or device that can have a plunger 602, a handle 604, a cavity 606 for holding a bioadhesive substance, an extender knob 608, a rod 612, a locking connector 616, and a flexible loop 620 configured to contain a film of bioadhesive substance. The loop 620 shown in FIG. 23 can optionally be configured to extend further than shown.

The plunger 602 may be used to expel a bioadhesive substance through a hollow cannula and/or into the flexible loop 620 when advanced distally on by a user (for example, pressure from a user's finger or using pneumatic pressure supplied by an external device connected to the handle). For

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example, the extender knob **608** may be used to extend the rod and, hence, the flexible loop **620** from a retracted position as seen in FIG. **23** to a more extended position or even a more retracted position. The knob **608** and rod **612** can be toothed such that a rotation of the knob in a first direction causes the teeth of the knob to engage the teeth of the rod to advance the rod in the distal direction. A rotation of the knob in a second direction causes the teeth of the knob to engage the teeth of the rod to retract the rod in the proximal direction. For example, the knob **608** can take the form of a roller wheel with teeth on a surface or with a smooth surface that can, when rolled, cause the movement of the rod either distally or proximally. The teeth can optionally be formed in a direction that is parallel with the rotational axis of the knob/roller wheel (not shown).

Alternatively, the knob and rod can be smooth, but in contact such that a rotation of the knob in a first direction causes the advancement of the rod and loop in the distal direction, and rotation of the knob in a second direction causes the retraction or withdrawal of the rod in the proximal direction. Any arrangements of the loop disclosed herein can be used to support a film of substance, such as a bioadhesive, that can be applied to an ocular tissue. This can be used with or without the introduction of an expansion fluid and, hence, with or without the formation of one or more bubbles using the loop. The loop can be, for example, placed in contact with or moved against the target tissue to deposit the substance carried by the loop on the target tissue.

Additionally, in any arrangements, the loop can be temperature controlled (i.e., cooled or heated) to change the adhesive or cohesive properties of film adherence of the substance to the loop, allow disengagement from the film, and/or allow stronger adhesion to the film. For example and without limitation, the wire can be heated or cooled with heating or cooling elements in communication with the loop to any desired temperature to promote release of the substance from the loop.

In some arrangements, bubbles can be formed of the substance retained by the loop. Additionally, in such arrangements, release of the bubble can be achieved by tightening or closing the loop (by withdrawing the loop, advancing a clasp, cinch, or other tightener to tighten the loop, or otherwise) to pinch the loop off of the bubble and detach the bubble from the device. The bubbles can be produced in a controlled fashion, such as being produced slowly, or in a controlled size. The bubbles can be produced one at a time. When a bubble is formed, it can be removed from the device onto a surface such as a retinal or other body tissue. The surface can be of a biological object including but not limited to an eye, a retina, a sclera, a conjunctiva, a hand, skin, or another body part.

FIG. **24** illustrates an example of a device **640** that can have a handle or housing **641**, an extender knob **648**, a rod **650**, a flexible loop **652** that can be configured to support a film of bioadhesive substance **654**, and two or more guide ridges **660** that can be used to guide the loop **652** through an opening **657** in the distal end of the device. The loop can be a retractable loop that can be advanced out of a distal end of the device, as shown in FIG. **25**. When the loop **652** is within the housing **641**, a substance can be applied to the loop **652** through an injection port, through a distal end of the device, through a passageway within the housing, and/or using any of the other device, components, or methods for advancing a substance through a cannula disclosed herein or known in the industry. A substance can also be added to the loop when the loop is positioned outside of the cannula.

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FIG. **25A** illustrates another example of an arrangement of a device **670** that can include a handle or housing **672**, a rod **674**, a cannula **676**, and a flexible loop **678** that can be configured to support a film of bioadhesive substance **680** across the loop **678**. The cannula may be detachable or non-detachable. Additionally, the loop **678** can optionally be detachable from the rod **674** so that the loop can remain in situ after the application of the substance to the target tissue surface. Alternatively, the loop **678** can be withdrawn back into the cannula **676** to cause the substance **680**, which can be in the form of a bubble or planar film, to release from the loop **678**. Note that the rod **674** can optionally extend from the handle portion, or proximal to the handle portion, so that a user can grasp and manipulate a proximal end **674a** of the rod **674** to advance and withdraw the rod **674** distally and proximally, respectively, rotate the rod **674** to manipulate the rod and loop **678**, or otherwise. In some arrangements, the rod can be keyed or indexed to the cannula to prevent the relative rotation of the rod relative to the cannula. The loop material in any embodiments described herein can include Nitinol, stainless steel, nylon, or any other desired polymeric or other acceptable material.

Additionally, in some arrangements, a desired substance can be applied to the loop **678** and the loop can be retracted within the cannula **676** before the device is delivered to the surgeon or the operating room. FIG. **25B** illustrates the device **670** with the rod **674** and, hence, flexible loop **678** withdrawn in a proximal direction relative to the handle **672** so that the flexible loop **678** is positioned at least partially (or completely, as shown) within the cannula **676**. For example and without limitation, the loop can be preloaded with any of a desired range of substances **680** and the loop **678** can be retracted within the cannula, as shown in FIG. **25B**. A plug or seal can be positioned over or into the cannula to prevent the inadvertent discharge or leakage of the substance from the device. In this preloaded state, the device **670** can be ready for use without the surgeon or user having to load the desired substance on the loop **678**. The seal can be removed and the rod **674** and loop **678** can be advanced past the distal end of the cannula once the device has been advanced into the desired location, wherein the substance can be deposited from the loop onto the target tissue or surface.

The rod of any arrangements disclosed herein (including but not limited to rod **612**, rod **650**, and/or rod **674**) can be hollow or can be surrounded by an additional sleeve, thereby providing an additional lumen (hereinafter referred to as a supply lumen) through which an additional or a first supply of a substance can be advanced to the loop. For example and without limitation, in some arrangements, the loop **678** (or any loop disclosed herein) can be withdrawn into the cannula and a supply of a substance can be advanced through the supply lumen, past a distal end **674b** of the rod **674** (or any rod disclosed herein) and onto the loop **678** while the loop is positioned within the cannula. This can result in the loop being loaded and/or reloaded with the substance within the cannula where the loop and the substance advanced through the supply lumen are confined within the cannula to prevent inadvertent contact of the substance to any unintended surfaces. Some arrangements of the system and/or device can include a trans-scleral tip. For example and without limitation, FIG. **26** illustrates an example of an arrangement of a cannula tip portion **690** for administering a substance, the device having cannula or body portion **691** and a trans-scleral tip **692**, among other features. The device **690** can also have a first sharp, pointed distal tip **692** that can be beveled or otherwise. The tip **692** can be configured to facilitate a penetration of the scleral. The device **690** can also

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have a distal port **696** that can be used for communication of a substance to a desired tissue surface. The port **696** can optionally include a rounded or smooth edge or bevel **698** around the distal port **696**. The port **696** may include a silicone diaphragm having any of the components and/or features of the diaphragm(s) disclosed above, or a diaphragm made out of another material. The end surface or tip portion **692** can be solid and can be beveled on both sides or on just one side, can have a conical shape, or otherwise. In some arrangements, the distal tip **692** can be sharp and can be used for cutting a tissue of the eye. The cannula tip **690** and its features can be used with any of the arrangements of devices disclosed herein.

Some arrangements of the system and/or device can include a tip plug. Non-limiting examples of tip plugs are shown in FIGS. **27** and **28**. The tip plugs **700**, **720** shown in FIGS. **27** and **28** can include a neck portion **702**, **722**, respectively. The neck portions **702**, **722** can be shaped to fit the dimensions of the tip of the cannula. For example, the neck portions **702**, **722** can be shaped to fit the dimensions of any substance retention rim, lip, or ring on or at the end or tip of the cannula. The neck portions **702**, **722** of the tip plugs **700**, **720** can include a beveled portion configured to be flush with a beveled portion of the distal tip. The tip plug can include a beveled portion configured to be flush with a beveled portion of the distal tip when the elongated member of the tip plug is placed within the distal tip.

The tip plugs **700**, **720** can include an elongated portion **703**, **723**, respectively. The elongated portion **703**, **723** of each plug can be connected to the neck **702**, **722** of each plug. The elongated portion **703**, **723** can include a proximal end and a distal end. The proximal end of the elongated portion **703**, **723** can be configured to fit within a distal tip of any arrangements of cannulas or devices disclosed herein. For example, the elongated portion **703**, **723** can fit inside a central tube of the system or device. The tip plug can include an elongated member comprising a proximal end and a distal end, wherein the proximal end of the elongated member can be configured to fit tightly (e.g., to prevent expansion media escape) within the distal tip. The elongated portion **703**, **723** can optionally be 1-5, 5-10, 10-20, 20-30, 30-40, or 40-50 mm in length. For example, as shown in FIG. **28**, the elongated portion can be formed in a range of lengths. The elongated member **1305**, **1405** can include an elongated plug tip that fits inside a central tube of the device or system, or a distal tip of the system or device.

The tip plugs **700**, **720** can include a tip plug handle or pull tab **704**, **724**. Tip plug handle **704**, **724** can be connected to the distal end of the elongated member **703**, **723** or to the neck **702**, **722**. For example, the tip plugs **700**, **720** can include a fingertip tab **704**, **724** that can be wider and more easily grasped compared to the elongated member **703**, **723**. The fingertip tab **704**, **724** may include a textured surface on one or more sides, such as ridges, dots, and/or hashes, or some other non-smooth surface. The tip plug can include a tab connected to the distal end of the elongated member of the tip plug.

With reference to FIG. **28**, the tip plug can include one or more supportive struts **726**. The supportive struts **726** can be configured prevent the tip plug **720** from breaking off when the elongated member **723** is inserted into the distal tip. The supportive struts **726** are configured to fit around the distal tip and can be have a continuous, annular shape, or two or more discrete struts. Some arrangements include a single circumferential supportive strut. The circumferential supportive strut can be an outer tube that fits over or around a portion of the tip of the cannula. The tip plug can include an

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outer tube that fits over or around a portion of the tip of the cannula. The elongated member **723** can be configured to fit inside the cannula. The central core can be longer, shorter, or the same length as the outer tube or supportive struts **726**. Some arrangements include a continuous, annular strut. Some arrangements include non-continuous struts.

As mentioned, some arrangements of the system and/or device include a tip cover such as a protective tip cover. A non-limiting example of another type of tip cover **730** that can be configured for use with any cannulas disclosed herein is shown in FIG. **29**. The tip cover **730** can include a first or proximal side and a second or distal side. The first side can optionally include an adhesive configured to engage the cannula. The protective tip cover can further include a nonadhesive tab, such as for example and without limitation, the nonadhesive tab **732** shown in FIG. **29**. The tab **732** can be configured to aid in removal of the tip cover **730** from the distal tip of a cannula. The tip cover can be removable when the system or device, or the distal tip of the system or device is ready to use. The cannula can be configured to be filled or contacted with a bioadhesive substance through a hub of the cannula or through tip of the cannula before securing any of the tip covers disclosed herein.

The tip plug or tip covers disclosed herein can be configured to protect the distal tip, maintain the sterility of the distal tip, or prevent the distal tip from getting dirty or cutting another item such as a fingertip or other object, and/or preventing a leakage of the substance contained within the cannula or device. For example, the tip plug or tip cover can be attached to the system or device during transportation of the device or system. The system or device can be preloaded with a substance such as a bioadhesive substance. The preloaded system or device can include the tip plug or tip cover. The tip plug or tip cover retains the bioadhesive substance within the preloaded system or device.

Any devices or arrangements disclosed herein can include one or more features configured to disengage the substance from the device. Such features can include, for example and without limitation, cutting features or components such as razor blades or sharp blades, thin metal or polymer blades, scrapers, spatulas or other similar devices configured to slide the substance off of the device. Additionally, in some arrangements, the bubble can be optionally clipped or pinched off the device.

Any arrangements of the systems and/or devices disclosed herein can include an internal cautery component. Some arrangements can be configured to have a cannula with a fiber or wire that can be used for light or cautery. FIG. **30** illustrates an example of an arrangement of a cannula **750** having a fiber or wire for light or cautery. The cannula **750** can include a light fiber **752** within a tube along a length of the cannula. The light fiber **752** can include an angled end portion **754** in the example, but, in other arrangements, the end of the light fiber **752** can be straight. The end portion **754** of the light fiber **752** can be beveled at an angle that matches the angle of the bevel of the distal tip.

The cannula **750** can include a light **756** emitted by the end **758** of the cannula **758** or light fiber **752**. Any arrangements of the devices disclosed herein can be configured to include one or more light fibers **752** or any of the other features of cannula **750**, and the cannula **750** can be configured to include any of the features of any of the other cannulas, devices, or other components or systems disclosed herein.

Additionally, an extra tube can be included that can be configured for passage of an additional material (such as a

liquid) that can be expressed from the distal tip as a spray or as drops. For example, the additional material may be used to activate the bioadhesive substance. The device or system, or a component thereof such as the handle, cannula, distal tip, light, or light fiber can be configured to emit a light at a wavelength that activates, thickens, or hardens the bioadhesive substance.

The system and/or device and/or components thereof may be activated in a manner that increases the temperature at the distal tip (using, for example, the internal cautery component). This can be advantageous according to some arrangements in aiding disengagement of the bioadhesive bubble from the distal tip in instances where the bioadhesive substance can be a temperature-responsive material that becomes more fluid and/or less cohesive at higher temperatures.

Some arrangements include a cryotherapy component such as a cryotherapy handpiece. The system or device can be activated in a manner that decreases the temperature at the distal tip (using, for example, a cryotherapy component). This can be advantageous according to some arrangements in aiding disengagement of the bubble or other form of the substance from the distal tip in instances where the bioadhesive substance can be a temperature-responsive material that becomes more fluid, friable, or releasable at lower temperatures.

Some arrangements of the systems and/or devices disclosed herein can have a special end portion that can have an increased application area, for example for applying heat or cold to a tissue surface or to a bioadhesive. Some arrangements can have a distal end portion having a solid circular shape, such as a button shape. FIG. 31 is a first perspective view and FIG. 32 is a second perspective view of an example of an end component 770 having a button tip arrangement. The button tip end component 770 can have an enlarged end surface or portion 772 and an elongated arm portion 774 coupled with the end portion 772 that can be straight or can be bent, as shown. The end portion 772 can be solid or can have one or more passageways therethrough, for expansion fluid, substance, or otherwise.

The button base can include a solid floor with an opening for the distal tip which ends flush with the button floor. The surface of the button can be open so as to form a reservoir configured to hold a substance such as a bioadhesive substance. The surface of the button can be solid with a central opening. The distal surface 776 of the end portion 772 can be generally planar, curved, or otherwise.

The button tip cannula can be configured to be heated or cooled. The button-tip cannula can be used to liquify a viscous, solidified, or partially solidified bioadhesive by holding the cooled or heated button-tip cannula over the bioadhesive substance until it can be converted into a liquid state enabling it to be aspirated with the same or a different device. The button tip design provides an area such as a broad area of either heat or cold over an area such as a broad area of bioadhesive providing more rapid liquification and removal of the bioadhesive than might otherwise be achieved with a cooling or heating tip with a smaller surface area.

Examples of substances that can be used with any of the devices, systems, and methods disclosed herein include, without limitation, adhesives, bioadhesives, gels, hydrogels, thick liquids or semi-liquid treatment substances, double layer hydrogels, nonsolids and the like. As used herein, any use of the term substance is intended to include any of the types and examples of substances disclosed anywhere in this disclosure.

Examples of substances that can be used with any of the devices, systems, and methods disclosed herein and which are contemplated as being included in each use of the term "substance" herein include, without limitation, any of the following: adhesives, bioadhesives, gels, hydrogels, thick liquids or semi-liquid treatment substances, double layer hydrogels, nonsolids, a polyethylene glycol solutions, a trilycine amine solutions, polymeric hydrogels, thermo-responsive gels, polyvinyl acetate, glue, aliphatic, cyanoacrylate, epoxy, polyurethane glue, and contact cement, glycoproteins, elastic, proteins, carbohydrates, mucopolysaccharides, hydrogels, double layer hydrogels, polymeric hydrogels, polyethylene glycol hydrogel and other hydrogels, biomimetic substances, ReSure (TM) sealant, a polyethylene glycol hydrogel, fibrin glue, a polyethylene glycol solution, a trilycine amine solution, glycoproteins, a polyethylene glycol solution, trilycine amine solution, elastic proteins, carbohydrates, mucopolysaccharides, temperature activated bioadhesives, UV or light activated bioadhesives, and vaccines, etc.

Any arrangements disclosed herein can also be configured to use one or more substances that polymerize as the temperature of the substance reaches a threshold value or range of values, such as for example and without limitation, normal body temperature, or to any values from approximately 80° F. to approximately 100° F. or greater, or from approximately 90° F. to approximately 100° F. or greater, or from approximately 95° F. to approximately 100° F. or greater, or to and from any values within these ranges. Additionally, any arrangements can be used to apply a cold cured substance to a tissue surface—i.e., substances that polymerize as the temperature of the substance drops to a threshold value or range of values, such as for example and without limitation, 10° F. below body temperature, or to any values from approximately 70° F. or less to approximately 90° F., or from approximately 80° F. to approximately 90° F., or to and from any values within these ranges.

Any arrangements disclosed herein can also be configured to use one or more substances that polymerize with activation by any number of various wavelengths of light, or are activated by light. Activation light may be provided by special applicator within or separate from the substance delivery device. Suitable substances that can be used also include substances that are typically used in or are approved for use in subretinal space, a vitreous cavity, or an ocular surface including on or under conjunctiva, and substances that may be used for closing gaps in tissue.

The substance can be configured so as to be safe and non-toxic to the eye, injectable, and bindable to a retinal tissue when hydrated and/or cured. The substance can also include substances with characteristics suitable for bubble/foam formation including but not limited to adhesive and non-adhesive substances for use both inside and outside the eye or any other human or animal tissue.

After creation of the bioadhesive bubble and dissociation of the bioadhesive bubble from the distal tip, the bioadhesive substance can be activated with application of a solution that may be delivered by the bioadhesive delivery device which may be constructed so that it has separate tubes for delivery of the bioadhesive substance and the activation solution.

The devices disclosed herein can be used for vaccine delivery, and the substance delivered by the device can include a vaccine. The vaccine delivery may comprise nasal delivery using a bubble, which may be a more efficient delivery than a spray delivery. The device may deliver the vaccine in a greater concentration over a greater surface area

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than existing vaccine delivery methods, potentially increasing the amount of vaccine absorbed by a subject to which the vaccine is applied.

Other substances that can be delivered with any of the arrangements disclosed herein and which are contemplated as part of the term substance, as used herein, include: substances that polymerize as they approach body temperature, substances that polymerize with activation by any number of various wavelengths of light, substances that are activated by light, and substances that are intended for use in a subretinal space, a vitreous cavity, or an ocular surface including on or under conjunctiva.

Any combination of the devices and/or components described herein can be provided together in a kit. Some arrangements of the kit further comprise a second system or device, and wherein the systems or devices are each disposable after a single or limited number of uses. Some arrangements of the kit further comprise an adhesive biomaterial. Some arrangements of the kit further comprise a substance dispenser. Some arrangements of the kit further comprise a tip plug or protective cover. The system, device, or kit described can be sterile or pre-sterilized. The kit can include a pre-sterilized substance dispenser described herein. The device can be used for non-medical substance delivery, including but not limited to adhesives, lubricants, insulators, sealants (for example, gas fittings, or pipe/plumbing fittings). This may be preferable to Teflon tape in hard to reach areas.

Arrangements of the system and/or device can comprise a substance dispenser. The substance dispenser can include a base with a well or a series of wells or open ended reservoirs. The dispenser can include a container made of plastic, metal or glass. The well or the series of wells can each comprise a substance such as the bioadhesive substance. Alternatively, the substance dispenser can include a series of distal tips preloaded with a substance configured to be selectively attached to a handpiece or cannula of any arrangements disclosed herein. The cannula or cannulas can be configured for a one-time or single use, or may be reusable. For example, the cannula can optionally be configured to be reusable within a period of time such as 1-60 min, 1-24 hours, and/or 1-3 days.

An example of a substance dispenser **790** is shown in FIGS. **33** and **34**. The example can include a base **792** having a series of wells or reservoirs **793** each comprising or preloaded with a bioadhesive substance or comprising a different cannula or tip portion. Each well can optionally contain a single dose of substance or multiple doses of a substance. The diameter of each well can be such that the cannula fits tightly within the well so that the substance may be efficiently and cleanly drawn into the cannula. In some arrangements, the substance dispenser **790** can have a plurality of cannulas that can be removed from the base **792** and attached to the device that is being used.

Some arrangements of the system and/or device include a method of using a substance dispenser. The method can include placing a cannula in a well containing a dose of substance, and/or optionally applying a small amount of gentle aspiration to the cannula to engage the substance within the tip of the cannula, and withdrawing the cannula from the dispenser. The wells of the substance dispenser can each include a tear-off cap or cover. The substance dispenser can be sterilized or pre-sterilized, and/or can include a sterilized liquid or substance such as a sterilized bioadhesive substance.

FIG. **35** is a section view of a distal tip **800** having a first and a second substance supply channel through at least a portion of the distal tip. An attachment or coupling **801** can

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be formed between an inner tube or channel **802** and an outer tube or channel **804**. The attachment may be along the entire length of the channels or may have intermittent attachments.

The distal tip **800** can have a space **808** between the inner tube **802** and the outer tube **804**. The distal tip **800** can also have a substance retention rim **810** extending radially inwardly adjacent to the distal end port or opening **812** of the distal tip **800**. The space provided inside the retention rim **810** can be filled with a substance and can aid in bubble formation. The example in FIG. **36** shows the attachments **30** between the inner tube **802** and the outer tube **804**. In some arrangements, connectors can be used to stabilize the inner tube **802** within the outer tube **804**.

Another arrangement of the device is shown in FIG. **37**. A first reservoir **812** (or a first substance supply channel) and a second reservoir **814** (or a second substance supply channel) can be formed along a length of the distal tip **800**. The first and second reservoirs can be divided and separate along the length thereof. In the example shown in FIG. **37**, the first reservoir can be configured to hold a first substance, and the second reservoir can be configured to remain empty and act as a receptacle for (or receive) a second substance or an activation solution as the second substance or activation solution advances out of the end of the first reservoir. A distal tip plug may be used to prevent efflux of substance out of the distal tip. The first and second reservoirs can each be configured to hold a separate bioadhesive substance or an activator substance, and/or the reservoirs can be configured to expel both bioadhesive substances together to allow the bioadhesive substances to mix and optionally react with each other prior to forming a film and bubble.

Any devices disclosed herein can be used to apply a bioadhesive bubble to a retina, comprising a proximal end comprising a handle, a distal end, an internal fluid passageway from the proximal end to the distal end, and a cannula connected to the distal end, comprising a distal tip, and configured to generate a bioadhesive bubble from a bioadhesive when a gas flows through the internal fluid passageway, maintain the bubble on the distal tip when the gas stops flowing through the internal fluid passageway, and release the bubble when the bubble can be brushed against a retina or when the gas flows again through the internal fluid passageway.

Some arrangements relate to a use of a system, device or kit for generating and/or applying the bioadhesive bubble to an eye or retina such as for repair of the eye or retina. The use can include generating a bioadhesive bubble using a system, device or kit described herein, and/or applying the bioadhesive bubble from the tip of the cannula to the eye or retina using a system, device or kit described herein.

Some arrangements relate to a method of repairing a retinal tear. Some arrangements of the method or use include generating, with a system, device or kit, a bioadhesive bubble comprising an approximately spherical film of bioadhesive substance enclosing an expansion fluid. Some arrangements of the method or use include applying the bioadhesive bubble to the eye or retina from the system, device or kit. Some arrangements relate to a method of repairing a retinal tear or hole, comprising: generating, with a system, device or kit, a bioadhesive bubble comprising an approximately spherical film of bioadhesive substance enclosing an expansion fluid, and applying the bioadhesive bubble to the eye or retina from the system, device or kit. Some arrangements include operation of a button on the handpiece of the device or system, or operation of a foot pedal to, for example, open a valve to allow expansion fluid

to flow through the expansion fluid passageway of the handpiece and/or through the fluid passageway of the cannula.

Some of the devices disclosed herein convert a cohesive liquid into a bubble with much larger surface area and control and accuracy of placement, than the original cohesive liquid without a bubble, significantly improving its capabilities when used for any of the following functions: closing gaps in tissue, reducing incidence of PVR by blocking liberation of RPE cells into the vitreous cavity.

The devices and methods disclosed herein can be used to close a variety of discontinuous and continuous structures including but not limited to retina tears and holes, macular holes, macular pits, break in posterior lens capsule, scleral laceration, corneal laceration, corneal abrasions, conjunctival laceration, retinal pigment epithelial tears, non-ocular medical uses include but are not limited to skin lacerations, mucosal laceration—regardless of location, and delivery of substance with no other functions, without limitation.

The substance delivered by any device disclosed herein can be used to deliver substance transvitreally. For transvitreal use, delivery through the device may employ a cutting action to amputate the cohesive/sticky substance once the substance has been advanced to the target location. Excess can be removable with a vitrectomy cutter and/or aspiration. Transvitreal delivery may be utilized in air filled eye to avoid adhesive substance floating to non-targeted ocular structures such as lens/IOL/pars plicata/angle, corneal endothelium which could result in obscuration of view through lens/IOL or impairment of inflow/outflow of aqueous humor.

Any devices disclosed herein can also be delivered by trans-scleral delivery, and may include delivery of substance to the retinal break through the subretinal space. After the substance is applied to the retina after being delivered via a tapered end of the device, or needle, the substance may be detached as the needle is withdrawn from the sclera. The substance may act as a buffer to prevent retinal incarceration into sclera defect left by the cannula, needle, or device utilized to cross the scleral and choroidal tissue.

Any arrangements of the delivery devices disclosed herein can have a single, double, or triple bore. The double bore device can include one bore for delivery of a substance and a second bore for reflux of intraocular fluid to outside the eye as IOP increases. One of the bores can contain a light fiber that may be used for illumination. One of the bores contains a fiber that can deliver a laser light. One of the bores contains a fiber that delivers heat or some other source of energy used for curing the injected substance. The triple bore device can include one bore for delivery of a substance, one bore for passage of gas used to create bubble or film of adhesive substance, and a third bore for reflux of intraocular fluid to outside the eye as IOP increases.

FIGS. 38-42 show schematic illustrations of examples of the distal end or tip of various arrangements 900, 910 of devices. In any arrangements, a single or dual compartment syringe can attach to a distal tip with dual compartments matching up when connected. With reference to FIG. 40 and FIG. 41 dual, isolated columns of air can optionally compress contents within the optional two distal tip compartments 912, 914. Some arrangements include a handle such as a syringe-like handle.

As shown in FIGS. 38-42, there may be an internal aspect of the distal tip, which may be flared to provide support and stability to a bubble at the tip. The tip may comprise material such as a hollow wall on the outside, and air or gas inside or in the internal aspect. The tip can have one or more

openings (such as passageways) that can communicate between hollow walls of the cannula containing material, and a central core of the cannula through which passes a gas such as air, nitrogen, oxygen, or another gas. A film of substance can be created across an opening in the distal end or distal tip. Hollow walls of the cannula may be of various volumes and relative volumes of hollow walls and the center core may vary. Some arrangements include a dual plunger for the central core, and/or a hollow walled outer cannula. The dual plunger may work and/or advance in unison or independently of the other components of the cannula or device. Components may be of various lengths, widths, and volumes.

In any arrangements disclosed herein, the device and/or method of use can be configured to have one or more of the following features, characteristics, capabilities, processes, or details: A hand held, syringe-shaped device that creates bubbles in controlled fashion, one bubble at a time; The ability to create bubbles from liquids of various compositions and viscosities; The ability to create bubbles from gels of various compositions and viscosities; The ability to maintain reservoir of substance, from which bubbles are created, at temperature within specific range depending on the substance used; A temperature-controllable chamber within device that can be calibrated specifically for the substance being used for task at hand so temp control can be on-off, though a temperature control mechanism could optionally be added to the device; A source of power comprising one or more batteries, wherein the batteries can optionally be incorporated into the hand-held device; A source of power comprising an electrical cord that connects to a vitrectomy machine power supply or to an external power supply; One or more color-coded devices with each color correlating with a specific substance composition, wherein the different substance compositions can optionally have a specific optimal temperature or temperature range to optimize bubble creation; A device configured to allow control and variation of bubble wall thickness based on, for example and without limitation, a temperature of the substance during bubble creation using one or more of heaters and/or coolers; A device configured to allow control and variation of bubble wall thickness based on the viscosity of the substance during bubble creation; A device configured to vary bubble wall thickness with various dimensions of certain device apertures and the amount of air or gas passed through the substance composition during the bubble making process, including without limitation a kit or system that has a plurality of different devices, each with a different distal tip aperture that generates bubbles with consistent size/diameter and wall thickness that is different from other devices of the kit; and another arrangement that has a device with an adjustable aperture and the ability to control a volume of the air injected to form the bubble, providing ability to customize size of bubble and thickness of bubble wall depending on clinical setting, for example in a retinal detachment application with multiple retinal tears of various sizes where it may be desirable to use bubbles or a variety of sizes but each having approximately the same bubble wall thickness.

In any arrangements disclosed herein, the device can be configured to also have one or more of the following features, characteristics, capabilities, or details: Using one device to close retinal tears and then at end of case to close external scleral opening through which cannulas were inserted (i.e. sclerotomies) in which case surgeon may wish to have access to greater bubble wall thickness due to greater tensile forces associated with scleral wound compared to

retinal break; Variations of the device to create bubbles of various sizes with various diameters ranging from, for example 0.5 mm-15 mm, other ranges of bubble diameters include, for example, 0.1 mm-0.5 mm, 0.5 mm-1 mm, 1 mm-2 mm, 2 mm-5 mm, 5 mm-10 mm, 10 mm-25 mm, 25 mm-50 mm, and 50 mm-100 mm, or any number therebetween; The ability to release the bubble and create additional bubbles that can also be released, which can be done in fashion that avoids plugging of cannula/device tip; A mechanism to release the bubble from cannula/device tip, which mechanism to release the bubble can include one or more of heating of tip to melt or heat the substance, chilling the tip to fracture the substance, and a device having a cutting or wiping mechanism at the tip; A device including a light fiber used to transmit light used for the curing of the substance used in the balloon, which light fiber can be fixed or it can be movable so as to be advanced out of the tip of the device, which light fiber may be advanced to the interior of the bubble to bring it into closer proximity to the portion of the bubble that is in contact with the retina, sclera, or other targeted human tissue, and/or which light fiber may be housed in a different bore of the cannula and be used to cure the substance from outside the balloon; and a device that may include a cannula and associated distal tip, separate from the bubble generating portion of device, that permits the dripping or spraying of a liquid or gel or other substance on the surface of retina prior to application of the bubble, wherein a purpose of this other substance may be to deposit a substance of specific composition that reacts with the bubble substance, resulting in a change in the consistency of both substances, for example from liquid to solid.

Any of the arrangements of the devices and systems disclosed herein can be used to apply an adhesive to a biological or non-biological surface or object, which object can include medical scaffolding, patches, covers, grafts, or other objects or implements used in medical and non-medical applications. For example and without limitation, FIG. 43 illustrates a non-limiting example of a system 1000 having a substance supply device 1001 and an applicator device 1002. The substance supply device 1001 can include any of the devices, arrangements, and/or components disclosed elsewhere herein in any combination to provide a substance to a surface or an object, including without limitation devices 100, 200, 250, 350, etc. The substance supply device 1001 can be advanced through a first introducer cannula or sheath into a desired space or location (such as, without limitation, an eye cavity that has been filled with air) and the applicator device 1002 can be advanced into the eye cavity through a second introducer cannula or sheath. The substance supply device 1001 can be used to apply a substance, which can include a bioadhesive, to a desired surface of the applicator device 1002.

FIGS. 44 and 45 depict a top view and a side view, respectively, of an example of the applicator device 1002 that can be used with any of the devices disclosed herein for supplying a substance to a surface or an object. As will be described, in any arrangements, the applicator device can be used to receive a substance on an applicator portion thereof for application of the substance to a tissue surface or an object, to support an object therewith (such as a patch or otherwise) and deliver the object to a desired location, or otherwise. For example and without limitation, the applicator device can be configured to support a patch, tissue cover, tissue graft, tissue scaffolding, or otherwise (collectively referred to herein as a patch) thereby, wherein the surgeon or user can use the applicator device 1002 to position and apply the patch to the desired target location. The patch can

optionally be pre-loaded with an adhesive (which can be a bioadhesive) or other substance, or can be loaded with the adhesive in the surgical location or space using the substance supply device 1001. Thereafter, once the patch has been loaded with the desired amount of adhesive, the patch can be applied to the desired object or tissue surface. Additionally, without limitation, the applicator device 1002 can be used to hold the patch in the desired application location to allow for the adhesive to sufficiently set or cure and, hence, bond to the target tissue.

With reference to FIGS. 43 and 44, the applicator device 1002 can include a cannula or outer sleeve 1003 (also referred to herein as an elongate body or outer sheath), an inner sleeve or rod 1005 that can optionally have a solid cross-section or a lumen or passageway 1007 extending therethrough, and an applicator tip 1006 coupled with the inner rod 1005. The outer sleeve 1003 can optionally be configured to provide additional rigidity and manipulability to the inner rod 1005 and at least a portion of the applicator tip 1006 to facilitate or improve the user's control of the applicator device 1002 during application of the substance 1001.

The inner rod 1005 can have a proximal end 1005a (not shown) and a distal end 1005b. The passageway 1007, if any, can extend from the proximal end 1005a to the distal end 1005b of the inner rod 1005. As shown, the applicator tip 1006 can be coupled with the distal end 1005b of the inner rod 1005. The proximal end of the inner rod 1005 can be manipulable by the surgeon or user of the device to advance, withdraw, rotate, and otherwise move the applicator tip 1006.

The cannula can optionally be flexible, semi-rigid, or rigid and can be formed from any suitable materials, including metal alloys, polymer materials, or otherwise. The system can optionally be configured to not have an outer sleeve 1003, wherein the inner rod 1005 and the applicator tip 1006 are sufficiently rigid and manipulable to not require the outer sleeve 1003.

In any arrangements disclosed herein, the applicator tip 1006 and/or other components can be pre-loaded in the cannula or outer sleeve 1003 such that the applicator tip 1006 need not be advanced into a proximal end of the cannula 1003 after the cannula 1003 has been advanced to the target location, and such that the size of the applicator device 1002 being advanced to the target location is defined by the size of the cannula 1003 and not the larger applicator tip 1006, thereby improving maneuverability of the device and reducing risk of injury the patient. Optionally, the applicator device 1002 can be configured such that the applicator tip 1006 is advanced into the proximal end of the cannula 1003 after the cannula 1003 has been advanced into the target location. A proximal end of the cannula 1003 can be flared or tapered to facilitate the insertion of the applicator portion 1010 of the applicator tip 1006 into the cannula 1003 and/or facilitate the collapse and/or rolling or furling of the applicator portion 1010 for insertion into the cannula 1003.

As mentioned, when the applicator tip 1006 is in the first or withdrawn state (i.e., within the cannula 1005), the distal portion of the applicator device 1002 can have a smaller profile or cross-sectional size which can be defined by the outer sleeve 1003 so as to be advanceable through an opening in a tissue surface with less force required and, consequently, less risk of trauma to the tissue. Some arrangements of the applicator device 1002 can be configured such that, when the applicator portion 1010 is withdrawn within or positioned within the outer sleeve 1003, the applicator

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portion 1010 can be biased to assume a curved, furled, folded, or rolled shape so as to fit within the inner space of the outer sleeve 1003, as best shown in FIG. 45.

With reference to FIG. 45, the applicator portion 1010 can be configured such that, when the inner rod 1005 and the applicator tip 1006 all withdrawn back into the outer sleeve 1003 from the second state to the first or withdrawn state, the applicator portion 1010 can be biased or otherwise configured to collapse to a more compact or narrow shape or state described above and shown in FIG. 45 so that the applicator device 1002 can be withdrawn through the opening in the tissue with the applicator tip 1006 withdrawn within the outer sleeve 1003. As shown and described, the applicator portion 1010 can have a width W that is greater than a width of the outer sleeve 1003. Because the applicator portion 1010 is configured or biased to move to a more compact profile upon withdrawal into the cannula 1003, withdrawing the applicator portion 1010 within the outer sleeve 1003 as shown in FIG. 45 can reduce the size of the distal portion of the applicator device 1002 to the size of the outer sleeve 1003, thereby reducing trauma to the patient during withdrawal of the device. In any arrangements, the applicator portion 1010 can be configured to have any features or be made from any materials that allow it to be folded, rolled, compressed, squeezed, narrowed or otherwise reduced in width W or size to fit within the outer sleeve when withdrawn within the outer sleeve.

The applicator portion 1010 can have or define a first width when the applicator portion 1010 is in the first position (wherein the applicator portion 1010 is contained within the outer sleeve 1003) and a second width when the applicator portion 1010 is in the second position (wherein the applicator tip extends past the opening at the distal end 1003a of the outer sleeve 1003). The second width of the applicator portion 1010 can be substantially greater than the first width of the applicator portion 1010. In any arrangements disclosed herein, the second width of the applicator portion 1010 can be approximately 5 times greater than the first width of the applicator portion 1010, or from approximately 2 times greater (or, optionally, less) to approximately 8 times greater (or, optionally, more) than the first width of the applicator portion 1010, or from approximately 3 times greater to approximately 6 times greater than the first width of the applicator portion 1010.

The inner rod 1005 can be advanced relative to the outer sleeve 1003 so that a distal end portion 1006a of the applicator tip 1006 can extend out of a distal end 1003a of the cannula 1003. In any arrangements, the applicator tip 1006 can have an elongate portion 1008 and an applicator portion 1010. The elongate portion 1008 of the applicator tip 1006 can optionally have a passageway 1012 extending therethrough from a proximal end 1008a to a distal end 1008b of the elongate portion 1008. The passageway 1012 can, in an operable state, be in fluid communication at the proximal end 1008a of the elongate portion 1008 with the passageway 1007, if any, of the inner rod 1005. One or, optionally, a plurality of openings 1014 can optionally be formed through a first surface 1011 of the applicator portion 1010, the openings or orifices 1014 being in fluid communication with the passageway 1012. The openings 1014 can be used to transmit a source of suction or negative pressure or a positive pressure or airflow to any objects in close proximity to the applicator portion 1010 or in contact with the applicator portion 1010. The source of suction can be communicated to the openings 1014 through the passageways 1007 and 1012, from, for example, a proximal end of the device such as through a handle coupled with the outer

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sleeve of the device. For example, a suction force can be applied through the openings to selectively attract and secure any desired objects against the first surface 1011 of the applicator portion 1010 including, without limitation, one or more bubbles comprising a substance (with, optionally, a gentle suction force to avoid bursting the bubble), a patch, tissue cover, scaffolding, or other object. For example and without limitation, a gentle suction force can be applied through the openings 1014 to attract and secure a bubble comprising a substance against the first surface 1011 of the applicator portion 1010.

Thereafter, the applicator portion 1010 can transfer the bubble to a desired location, such as a tissue defect in a surgical airspace within an eye. The substance can be released from the applicator portion 1010 by reducing the suction force or negative pressure provided through the openings 1014, by pressing and/or manipulating the applicator portion 1010 against the target surface, and/or by exerting a positive pressure through the openings, such as optionally a short burst or positive pressure. A supply of positive pressure (e.g., through a provision of a positive airflow through the opening or openings 1014) can be used to facilitate the release of the bubble.

In another nonlimiting example, negative pressure can be used to selectively secure a patch or any other implement against the first surface 1011 of the applicator portion 1010. For example, as shown in FIG. 46, a source of suction can be applied through the openings 1014 to a patch or other medical implement 1025 to selectively and releasably hold the patch 1025 against the first surface 1011 of the applicator portion 1010. Thereafter, the patch 1025 can be delivered to a desired location and advanced into contact with the target tissue surface by advancing the applicator device 1002 to the target location. When the desired position of the patch 1025 has been achieved, and the patch has optionally been retained in position long enough to secure the patch to the target tissue surface, the negative pressure may be reduced and the patch released from the applicator portion 1010. A positive pressure or airflow can optionally be provided through the openings to facilitate the release of the patch.

Additionally, with reference to FIG. 46, a source of suction can be applied through the openings 1014 to a patch or other medical implement 1025 to selectively and releasably hold the patch 1025 against the first surface 1011 of the applicator portion 1010. Thereafter, the substance supply device 1001 (which can be or have any of the components or details of any of the arrangements of the bubble, loop, or other substance supply devices disclosed herein) can provide a bubble 146 or layer of substance 105 on a surface of the patch 1025. The substance 105 can be a bioadhesive. The bubble 146 and/or the substance 105 can be applied against the patch as described above for device 130 in FIGS. 4A-4D. After the substance 105 has been applied to the patch, the patch 1025 can be delivered to a desired location and the adhesive coated surface of the patch 1025 can be advanced into contact with the target tissue surface by advancing the applicator device 1002 to the target location, as shown in FIG. 47. When the desired position of the patch 1025 has been achieved, and the patch has optionally been retained in position long enough to secure the patch to the target tissue surface, the negative pressure may be reduced and the patch released from the applicator portion 1010. In any arrangements, the applicator portion 1010 may be biased and configured such that, as the applicator portion 1010 is withdrawn into the cannula 1003, interaction with the distal end of the cannula 1003 will cause the applicator portion

1010 to assume a folded or furled shape so that the applicator portion **1010** can be completely withdrawn into the cannula.

In other arrangements, the applicator portion **1010** can be solid and without any openings or orifices therein. Additionally, the applicator tip **1006** can have a planar shape, a curved shape as shown in FIG. **46**, or otherwise. In other arrangements, clips, tabs, depressions, ridges or protrusions, or other features can be used to selectively and releasably retain the patch or other medical implement to the first surface **1011** of the applicator portion **1010**. In other arrangements, the applicator portion **1010** and applicator device **1002** can be configured so that a patch, cover, or other medical implement can rest freely on the first surface **1011** of the applicator portion **1010**. In any arrangements, the patch can come preloaded in the cannula **1002**, and can be rolled up against the surface **1011** of the applicator portion **1010** so that, as the applicator portion **1010** is advanced out of the distal end of the cannula **1003**, the applicator portion **1010** can self-expand or open up to an open shape, with the patch **1025** still retained against the first surface **1011** of the applicator portion **1010**.

In any arrangements, the patch can be secured at a proximal end of the patch using end-grasping forceps that are extendable from a cannula. The forceps can be self-expanding such that, as the cannula is retracted relative to the forceps or the forceps are advanced distally relative to the distal end of the cannula, the forceps will open up and release the grip on the patch. In this arrangement, the patch can be preloaded inside a distal end of the cannula with at least a portion of the patch grasped by the forceps. Because the forceps are constrained by the cannula, the forceps will retain their grasp on the patch until the forceps are advanced distally out of the cannula or the cannula is withdrawn proximally relative to the forceps. For example, as the forceps are advanced distally relative to the cannula, the patch can be moved past the distal end of the cannula and allowed to unfold, unroll, or unfurl into a relaxed, expanded state while the forceps are still sufficiently constrained in the closed position by the cannula such that the forceps continue to grasp the patch. In this state and position, a substance such as a bioadhesive can be applied to a desired surface of the open or relaxed patch using any of the substance applicator devices disclosed herein, for example and without limitation device **130**. The substance can be applied as a bubble, a plurality of bubbles, or otherwise. The substance applicator can also optionally be used to spread the substance over the surface of the patch. After the adhesive has been sufficiently applied to the surface of the patch, the patch can be positioned in contact with the desired tissue surface until the substance has created a sufficient bond between the patch and the tissue surface. Thereafter, the forceps can be further expanded to release the grasp of the forceps on the patch by advancing the forceps distally relative to the cannula and/or retracting the cannula to permit the forceps to expand, by button or slide action, or otherwise. Thereafter, the forceps can be withdrawn back into the cannula and withdrawn from the surgical site.

The forceps or other grasping means disclosed in any arrangements herein can be or include the ALCON Revolution DSP forceps or GRIESHABER DSP forceps. For example, such forceps can be used to grasp and/or manipulate a patch used in any arrangements disclosed herein.

The patch can also be released from the applicator portion using any mechanical components, including a button, that a surgeon or user can advance to advance the end-grasping forceps and attached patch and/or patch material. If the

forceps are loaded into the passive-open position, then once the forceps are extended past the distal tip of the cannula, the forceps can be biased or configured to open or return into their open or relaxed state.

In another method, the same internal end-grasping forcep can pinch the patch in its tips, but use a button in the reverse manner—to retract the cannula from around the end-grasping forceps and attached patch. This can allow surgeon to hold the tip of the cannula close to the target tissue and slowly expose the patch without actually advancing the patch forward, thereby avoiding hitting the retina with the patch which can occur if the surgeon does not compensate by gradually pulling the cannula back away from retina as tip advances. Once the cannula is fully retracted proximal to the tips of the end-grasping forceps, the forceps would be biased to passively open and release the patch or spatula or both.

Additionally, some arrangements can have a selectively disconnectable connector **1022** that couples the proximal end **1008a** of the elongate body **1008** with the distal end **1005b** of the inner rod **1005**. In some arrangements, the applicator tip **1006** including, hence, at least the applicator portion **1010**, can be released from the inner rod **1005** by disengaging the connector **1022**. This can allow the applicator tip **1006** to be removed from the device and left in situ within the patient's body. In this configuration, the applicator portion **1010** can essentially be the patch or cover for the defect. The applicator tip **1006** can be made from any suitable material, including any suitable bioabsorbable or biocompatible materials.

In any arrangements of the applicator device **1002** or any substance supply device embodiments disclosed herein, the handle, cannula or outer sheath **1003**, the inner sheath **1005**, or other rigid or semi-rigid components can be made from or comprise any suitable plastic material, patch material, metal, polyvinyl chloride, glass, acrylic, carbon fibers and/or any combination of the foregoing. In some arrangements, the cannula tip plug or protective tip cover can comprise plastic, metal, polyvinyl chloride, glass, acrylic, carbon fibers, rubber (such as, without limitation, silicon), and/or any combination of the foregoing.

While certain embodiments of the inventions have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of the disclosure. Indeed, the novel methods and systems described herein may be embodied in a variety of other forms. Furthermore, various omissions, substitutions and changes in the systems and methods described herein may be made without departing from the spirit of the disclosure. The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the disclosure. Accordingly, the scope of the present inventions is defined only by reference to the appended claims.

Features, materials, characteristics, or groups described in conjunction with a particular aspect, embodiment, or example are to be understood to be applicable to any other aspect, embodiment or example described in this section or elsewhere in this specification unless incompatible therewith. All of the features disclosed in this specification (including any accompanying claims, abstract and drawings), and/or all of the steps of any method or process so disclosed, may be combined in any combination, except combinations where at least some of such features and/or steps are mutually exclusive. The protection is not restricted to the details of any foregoing embodiments. The protection extends to any novel one, or any novel combination, of the

features disclosed in this specification (including any accompanying claims, abstract and drawings), or to any novel one, or any novel combination, of the steps of any method or process so disclosed.

Furthermore, certain features that are described in this disclosure in the context of separate implementations can also be implemented in combination in a single implementation. Conversely, various features that are described in the context of a single implementation can also be implemented in multiple implementations separately or in any suitable subcombination. Moreover, although features may be described above as acting in certain combinations, one or more features from a claimed combination can, in some cases, be excised from the combination, and the combination may be claimed as a subcombination or variation of a subcombination.

Moreover, while operations may be depicted in the drawings or described in the specification in a particular order, such operations need not be performed in the particular order shown or in sequential order, or that all operations be performed, to achieve desirable results. Other operations that are not depicted or described can be incorporated in the example methods and processes. For example, one or more additional operations can be performed before, after, simultaneously, or between any of the described operations. Further, the operations may be rearranged or reordered in other implementations. Those skilled in the art will appreciate that in some embodiments, the actual steps taken in the processes illustrated and/or disclosed may differ from those shown in the figures. Depending on the embodiment, certain of the steps described above may be removed, others may be added. Furthermore, the features and attributes of the specific embodiments disclosed above may be combined in different ways to form additional embodiments, all of which fall within the scope of the present disclosure. Also, the separation of various system components in the implementations described above should not be understood as requiring such separation in all implementations, and it should be understood that the described components and systems can generally be integrated together in a single product or packaged into multiple products.

For purposes of this disclosure, certain aspects, advantages, and novel features are described herein. Not necessarily all such advantages may be achieved in accordance with any particular embodiment. Thus, for example, those skilled in the art will recognize that the disclosure may be embodied or carried out in a manner that achieves one advantage or a group of advantages as taught herein without necessarily achieving other advantages as may be taught or suggested herein.

Conditional language, such as “can,” “could,” “might,” or “may,” unless specifically stated otherwise, or otherwise understood within the context as used, is generally intended to convey that certain embodiments include, while other embodiments do not include, certain features, elements, and/or steps. Thus, such conditional language is not generally intended to imply that features, elements, and/or steps are in any way required for one or more embodiments or that one or more embodiments necessarily include logic for deciding, with or without user input or prompting, whether these features, elements, and/or steps are included or are to be performed in any particular embodiment.

Conjunctive language such as the phrase “at least one of X, Y, and Z,” unless specifically stated otherwise, is otherwise understood with the context as used in general to convey that an item, term, etc. may be either X, Y, or Z. Thus, such conjunctive language is not generally intended to

imply that certain embodiments require the presence of at least one of X, at least one of Y, and at least one of Z.

Language of degree used herein, such as the terms “approximately,” “about,” “generally,” and “substantially” as used herein represent a value, amount, or characteristic close to the stated value, amount, or characteristic that still performs a desired function or achieves a desired result. For example, the terms “approximately,” “about,” “generally,” and “substantially” may refer to an amount that is within less than 10% of, within less than 5% of, within less than 1% of, within less than 0.1% of, and within less than 0.01% of the stated amount. As another example, in certain embodiments, the terms “generally parallel” and “substantially parallel” refer to a value, amount, or characteristic that departs from exactly parallel by less than or equal to 15 degrees, 10 degrees, 5 degrees, 3 degrees, 1 degree, or 0.1 degree.

The scope of the present disclosure is not intended to be limited by the specific disclosures of preferred embodiments in this section or elsewhere in this specification, and may be defined by claims as presented in this section or elsewhere in this specification or as presented in the future. The language of the claims is to be interpreted broadly based on the language employed in the claims and not limited to the examples described in the present specification or during the prosecution of the application, which examples are to be construed as non-exclusive.

What is claimed is:

1. A device for applying a bubble of a bioadhesive substance to a tissue surface of an eye, the device comprising:
 - a cannula comprising a proximal end portion, a distal end portion, and an intermediate portion extending therebetween;
 - a distal tip at the distal end portion of the cannula configured for insertion into the eye, the distal tip having an outer, longitudinally extending surface, the distal tip having a wall with an inner surface defining a lumen of a first diameter for passage of an expansion gas, the wall having a thickness between the distal tip outer surface and the wall inner surface, the distal tip defining an exit port in fluid communication with the lumen, the exit port having a diameter that is smaller than the lumen first diameter, the exit port circumvented by a rim that extends from the distal tip outer surface inward relative to the wall inner surface, the rim having a flat, circumferential face forming a distal most surface of the distal tip, an entirety of the circumferential face being flat, the flat, circumferential face having a width between an outer diameter and an inner diameter of the flat, circumferential face that is larger than the wall thickness, the rim configured to support a layer of the bioadhesive substance on the flat, circumferential face;
 - a bioadhesive pathway for introducing the bioadhesive substance to the flat, circumferential face of the rim;
 - an expansion gas passageway extending through at least the intermediate portion and the distal end portion of the cannula, the expansion gas passageway being in fluid communication with the distal tip lumen and being configured to provide a passageway for the expansion gas separate from the bioadhesive pathway along a length of the intermediate and distal end portions of the cannula all the way to the exit port, the expansion gas passageway being coaxial with the bioadhesive pathway; and
 - an actuator configured to couple with a source of the expansion gas and configured to selectively advance

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the expansion gas through the expansion gas passageway and the distal tip lumen upon actuation of the actuator and to direct the expansion gas to the layer of the bioadhesive substance to form the bubble of the bioadhesive substance on the flat, circumferential face of the rim with the layer of the bioadhesive substance at least partially surrounding a volume of the expansion gas.

2. The device of claim 1, comprising a handle portion at a proximal end of the device, the handle portion configured to support at least the actuator and the source of expansion gas.

3. The device of claim 1, wherein the device is configured to form a bubble that comprises a spherically shaped film of the bioadhesive substance at least partially enclosing the expansion gas.

4. The device of claim 1, wherein the actuator comprises a compressible bladder configured to expel the expansion gas from the source of the expansion gas within the bladder through the expansion gas passageway.

5. The device of claim 1, wherein the actuator comprises a wheel moveable along a compressible bladder configured to expel the expansion gas from the source of the expansion gas within the bladder through the expansion gas passageway.

6. The device of claim 1, wherein the bioadhesive pathway surrounds the expansion gas passageway.

7. The device of claim 1, wherein the bioadhesive pathway comprises an elongate body comprising a proximal end and a distal end having a distal tip with at least one opening therein, the elongate body being advanceable through the expansion gas passageway to the distal tip at the distal end portion of the cannula.

8. The device of claim 1, wherein the bioadhesive pathway is internal to or surrounded by at least a portion of the expansion gas passageway of the cannula.

9. The device of claim 1, further comprising a cautery component configured to increase the temperature of at least the distal tip of the device.

10. The device of claim 1, wherein the flat, circumferential face of the rim is not beveled perpendicular to a longitudinal extent of the cannula.

11. The device of claim 1, wherein the flat, circumferential face of the rim is beveled.

12. The device of claim 1, wherein the width of the flat, circumferential face is in the range of 0.1 mm to 0.5 mm.

13. The device of claim 1, wherein the width of the flat, circumferential face is in the range of 0.5 mm to 1 mm.

14. The device of claim 1, wherein the distal tip defines a single exit port.

15. The device of claim 1, further comprising a manifold surrounding the exit port, the bioadhesive pathway being in fluid communication with the manifold.

16. The device of claim 1, wherein the distal tip comprises a hydrophobic material.

17. The device of claim 1, wherein the rim is configured to support the layer of the bioadhesive substance on an entirety of the flat, circumferential face.

18. A handheld device for applying a bioadhesive bubble to a retina of an eye, comprising:

an elongate body comprising a first end and a second end, the first end comprising a handle, the second end comprising a cannula configured for insertion into the eye and having a wall with an outer, longitudinally extending surface, the cannula defining a lumen and having an exit port and a rim having a flat face at a distal most surface of the cannula, an entirety of the rim

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face being flat, the exit port having a smaller diameter than the lumen and the flat face extending inward from the outer surface into the lumen a distance greater than a thickness of the cannula wall adjacent the flat face; a bioadhesive reservoir configured for containing a liquid bioadhesive;

a first passageway in communication with the exit port and configured to selectively communicate a supply of an expansion gas through the exit port; and

a second passageway in communication with the bioadhesive reservoir and configured to selectively communicate a supply of the bioadhesive from the bioadhesive reservoir to the second end of the elongate body, wherein the second passageway completely surrounds the first passageway at least at a distal end of the cannula all the way to the exit port;

wherein:

the first passageway is configured to direct the supply of the expansion gas through a layer of the bioadhesive that is received on the flat face to produce the bioadhesive bubble;

the first passageway is configured to maintain the supply of the expansion gas out of contact with the bioadhesive until the expansion gas is directed through the bioadhesive at the flat face;

the flat face is configured to maintain the bioadhesive bubble on the second end when the expansion gas stops flowing through the exit port; and

the second end is configured to release the bioadhesive bubble when the bioadhesive bubble is brushed against a retina or when the expansion gas flows again through the exit port.

19. The handheld device of claim 18, wherein the elongate body further comprises a joining portion, the joining portion comprising a coupling end configured to interact with a proximal portion of the cannula, and wherein the proximal portion of the cannula is configured to reversibly interact with the coupling end of the elongate body.

20. The device of claim 18, wherein the flat face has a width in the range of 0.1 mm to 0.5 mm.

21. The device of claim 18, wherein the flat face has a width in the range of 0.5 mm to 1 mm.

22. The device of claim 18, wherein the distal tip defines a single exit port.

23. The device of claim 18, further comprising a manifold surrounding the exit port, the bioadhesive pathway being in fluid communication with the manifold.

24. The device of claim 18, wherein the distal tip comprises a hydrophobic material.

25. A device for applying a bubble of a bioadhesive substance to a tissue surface, the device comprising:

a cannula comprising a proximal end portion, a distal end portion, and an intermediate portion extending therebetween along a longitudinal axis;

a distal tip at the distal end portion of the cannula, the distal tip having an outer longitudinally extending surface, the distal tip having a wall of a first thickness defining a lumen of a first diameter for passage of an expansion gas, the distal tip defining only a single exit port, the exit port having a smaller diameter than the lumen first diameter, the exit port circumvented by a rim of larger dimension than the first thickness, the rim having a circumferential face extending inward from the distal tip outer surface, the circumferential face oriented at an acute angle to the longitudinal axis along

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an entirety of the circumferential face, the rim configured to support a layer of the bioadhesive substance on the circumferential face;

a bioadhesive pathway for introducing the bioadhesive substance to the circumferential face of the rim;

an expansion gas passageway extending through at least the intermediate portion and the distal end portion of the cannula, the expansion gas passageway being in fluid communication with the distal tip lumen and being configured to provide a passageway for the expansion gas separate from the bioadhesive pathway along a length of the intermediate and distal end portions of the cannula, the expansion gas passageway and bioadhesive pathway remaining separate all the way to the exit port; and

an actuator configured to couple with a source of the expansion gas and configured to selectively advance the expansion gas through the expansion gas passageway and the distal tip lumen upon actuation of the actuator and to direct the expansion gas to the layer of the bioadhesive substance to form the bubble of the bioadhesive substance on the circumferential face of the rim with the layer of the bioadhesive substance at least partially surrounding a volume of the expansion gas.

26. The device of claim 25, wherein the circumferential face has a width in the range of 0.1 mm to 0.5 mm.

27. The device of claim 25, wherein the circumferential face has a width in the range of 0.5 mm to 1 mm.

28. The device of claim 1, further comprising a manifold surrounding the exit port, the bioadhesive pathway being in fluid communication with the manifold.

29. The device of claim 25, wherein the distal tip comprises a hydrophobic material.

30. The device of claim 25, wherein the distal tip is configured for insertion into the eye.

31. The device of claim 25, wherein the rim is configured to support a layer of the bioadhesive substance on the entirety of the circumferential face.

32. The device of claim 25, wherein an entirety of the circumferential face is flat.

33. A device for applying a bubble of a bioadhesive substance to a tissue surface, the device comprising:

a cannula comprising a proximal end portion, a distal end portion, and an intermediate portion extending therebetween along a longitudinal axis;

a distal tip at the distal end portion of the cannula, the distal tip having a wall of a first thickness defining a lumen of a first diameter, the distal tip defining an exit port in fluid communication with the lumen, the exit port having a smaller diameter than the lumen first diameter, and a deflector extending into the lumen and having a distal facing surface located within the distal tip;

a bioadhesive pathway for introducing the bioadhesive substance to the distal facing surface of the deflector;

an expansion gas passageway defined by a member extending through at least the intermediate portion and the distal end portion of the cannula, the expansion gas passageway being in fluid communication with the lumen and being configured to provide a passageway for the expansion gas separate from the bioadhesive pathway along a length of the intermediate and distal end portions of the cannula, the deflector configured to direct flow of the bioadhesive substance radially inwardly about a distal end of the member defining the

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passageway for the expansion gas, the distal end of the member spaced proximal of the deflector; and

an actuator configured to couple with a source of the expansion gas and configured to selectively advance the expansion gas through the expansion gas passageway and the lumen upon actuation of the actuator and to direct the expansion gas to a layer of the bioadhesive substance to form the bubble of the bioadhesive substance on the distal facing surface of the deflector with the layer of the bioadhesive substance at least partially surrounding a volume of the expansion gas.

34. The device of claim 33, wherein the distal tip is configured for insertion into the eye.

35. A device for applying a bubble of a bioadhesive substance to a tissue surface of an eye, the device comprising:

a cannula comprising a proximal end portion, a distal end portion, and an intermediate portion extending therebetween;

a distal tip at the distal end portion of the cannula configured for insertion into the eye, the distal tip having an outer, longitudinally extending surface, the distal tip having a wall with an inner surface defining a lumen of a first diameter for passage of an expansion gas, the wall having a thickness between the distal tip outer surface and the wall inner surface, the distal tip defining an exit port in fluid communication with the lumen, the exit port having a diameter that is smaller than the lumen first diameter, the exit port circumvented by a rim that extends from the distal tip outer surface inward relative to the wall inner surface, the rim having a flat, circumferential face forming a distal most surface of the distal tip, an entirety of the circumferential face being flat, the flat, circumferential face having a width between an outer diameter and an inner diameter of the flat, circumferential face that is larger than the wall thickness, the rim configured to support a layer of the bioadhesive substance on the flat, circumferential face;

a bioadhesive pathway for introducing the bioadhesive substance to the flat, circumferential face of the rim;

an expansion gas passageway having a central axis extending through at least the intermediate portion and the distal end portion of the cannula, the expansion gas passageway being in fluid communication with the distal tip lumen and being configured to provide a passageway for the expansion gas separate from the bioadhesive pathway along a length of the intermediate and distal end portions of the cannula all the way to the exit port, the central axis of the expansion gas pathway being aligned with a center of the exit port; and

an actuator configured to couple with a source of the expansion gas and configured to selectively advance the expansion gas through the expansion gas passageway and the distal tip lumen upon actuation of the actuator and to direct the expansion gas to the layer of the bioadhesive substance to form the bubble of the bioadhesive substance on the flat, circumferential face of the rim with the layer of the bioadhesive substance at least partially surrounding a volume of the expansion gas.

36. The device of claim 35, wherein the flat face has a width in the range of 0.1 mm to 0.5 mm.

37. The device of claim 35, wherein the flat face has a width in the range of 0.5 mm to 1 mm.

38. The device of claim 35, wherein the distal tip defines a single exit port.

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39. The device of claim **38**, wherein the distal tip defines only the single exit port.

40. The device of claim **35**, further comprising a manifold surrounding the exit port, the bioadhesive pathway being in fluid communication with the manifold.

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41. The device of claim **35**, wherein the distal tip comprises a hydrophobic material.

42. The device of claim **35**, wherein the rim is configured to support the layer of the bioadhesive substance on an entirety of the flat, circumferential face.

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